

Election Irregularity Detection System

By

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The Kelkar Education Trust's

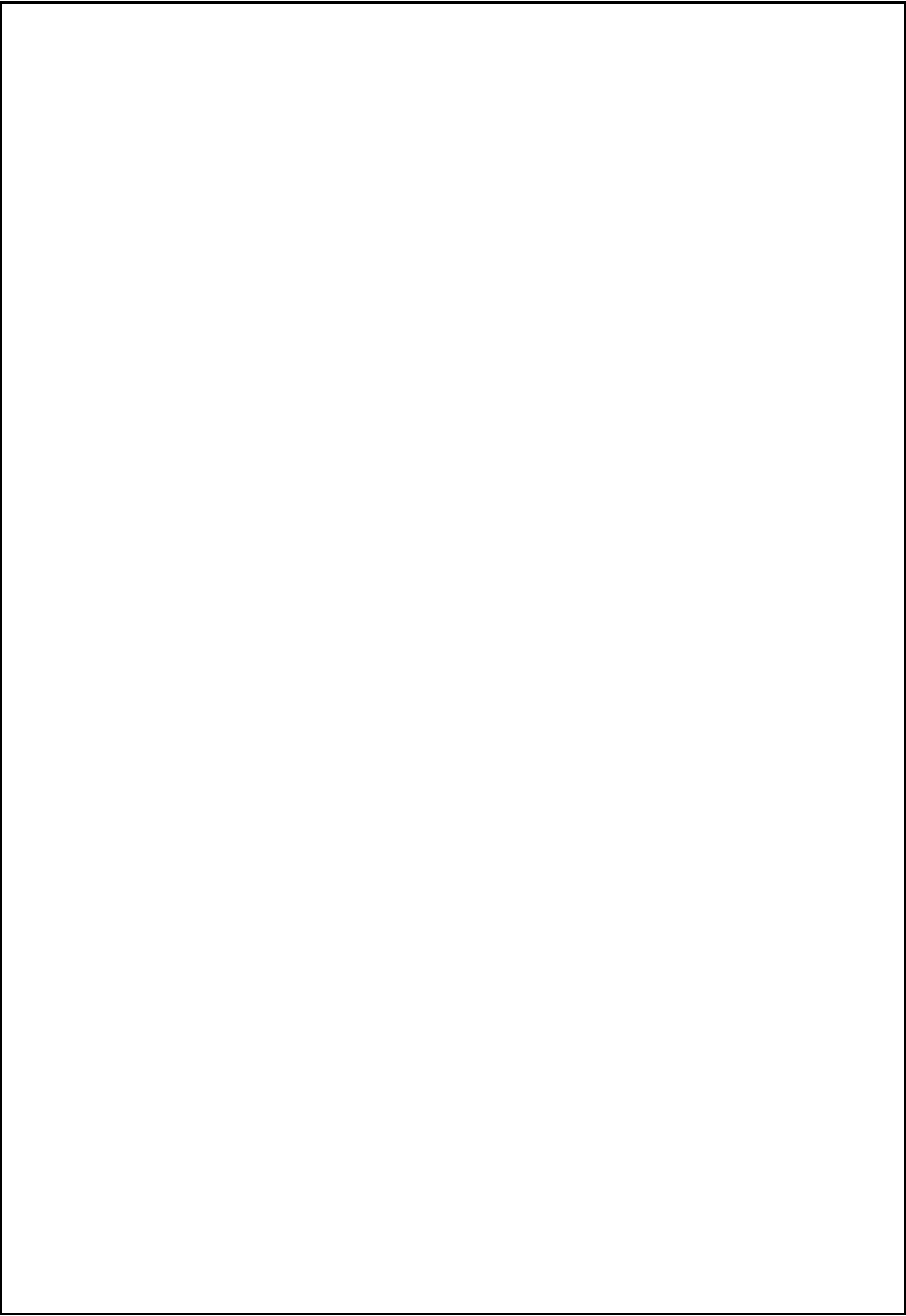
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Declaration

I hereby declare that the project titled **“Election Irregularity Detection System”** submitted in partial fulfillment of the requirements for the Master of Science in Information Technology is my original work.

This project has been conducted under the guidance of Mrs. Pournima Bhangale and has not been submitted earlier, in part or in full, for the award of any other degree or diploma at any university or institution.

I further declare that this project is free from plagiarism and has been prepared in accordance with the university's ethical guidelines and academic standards. Any references, citations, or works of others have been properly acknowledged and cited in this project.

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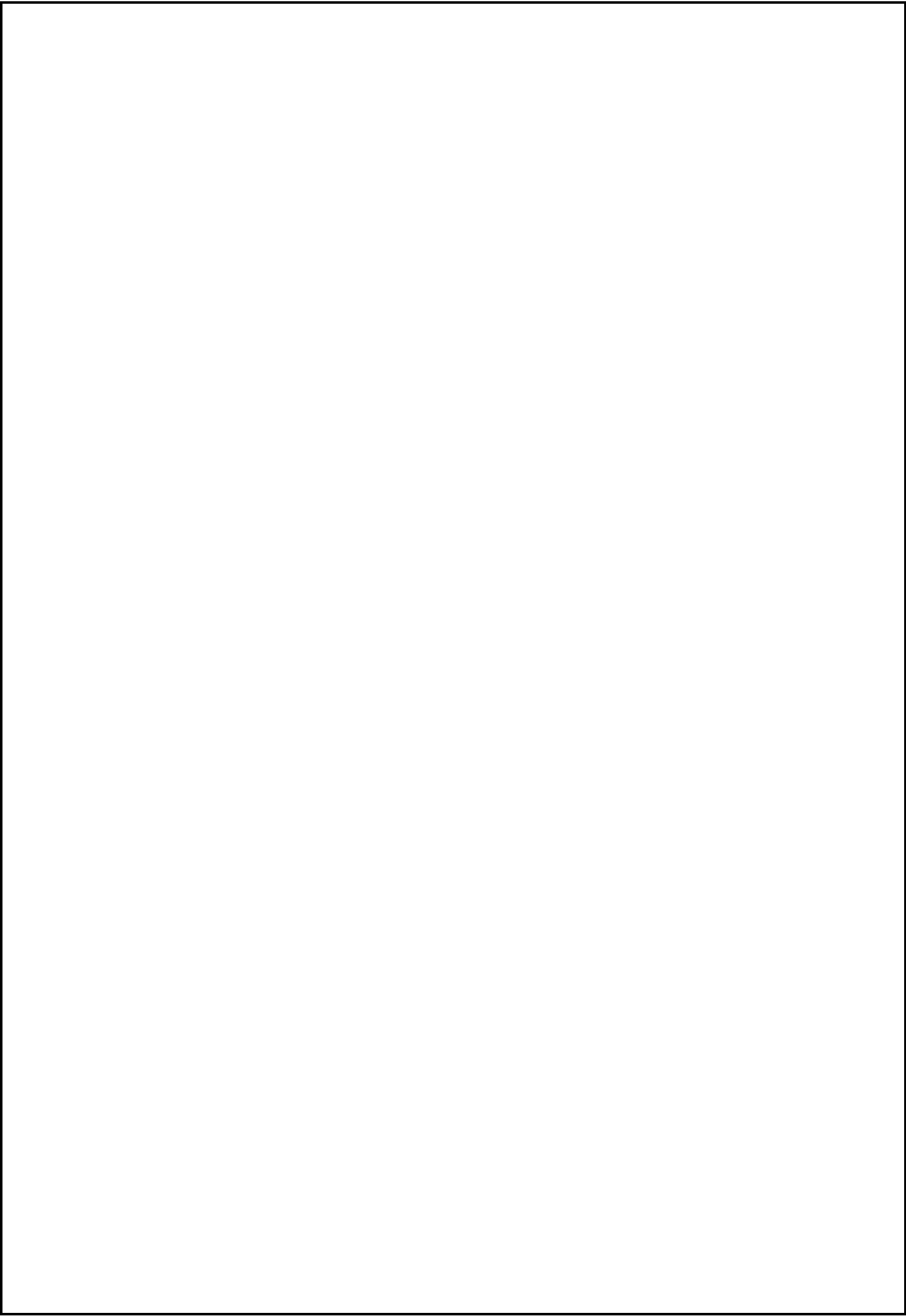
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ABSTRACT

Elections are the foundation of democratic governance, and ensuring their integrity is one of the most critical responsibilities of any democratic institution. With the increasing availability of digitized electoral data, there exists a significant opportunity to apply data-driven methods to the problem of electoral irregularity detection — moving beyond reactive, complaint-driven oversight toward proactive, systematic, and evidence-based monitoring.

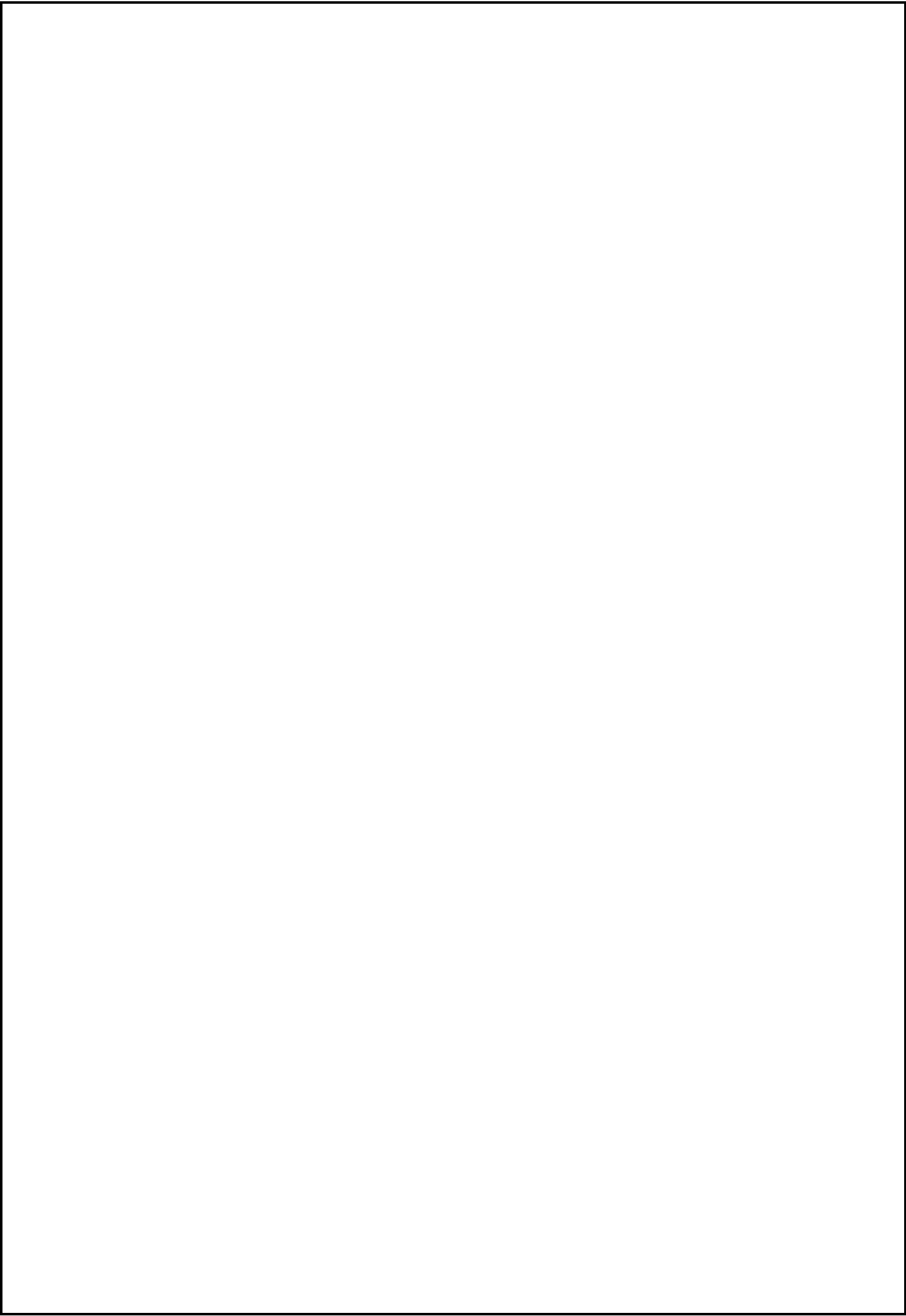
This project presents the design, development, and evaluation of a **Machine Learning-Based Election Irregularity Detection System** — an end-to-end analytical pipeline that processes historical polling booth-level election data to automatically identify statistically anomalous voting patterns and classify them by risk level. The system is built entirely using open-source Python tools and is deployed through an interactive Streamlit dashboard accessible to non-technical users.

The dataset used in this project comprises **73,081 constituency-level records** spanning **11 Lok Sabha election cycles from 1977 to 2014**, covering **43 states and union territories** across India. The raw data is processed through a systematic preprocessing pipeline that handles missing values, removes duplicates, and validates data integrity. Five contextual features are then engineered from the cleaned data — Turnout Percentage, Turnout Change, State Average Turnout, State Deviation, and Normalized Turnout — each capturing a distinct dimension of electoral behavior at the booth level.

Two complementary unsupervised machine learning algorithms form the core of the detection engine. **Isolation Forest** detects global anomalies by measuring how easily each data point can be isolated through random partitioning of the feature space. **Local Outlier Factor (LOF)** detects contextual anomalies by comparing the local density of each data point to that of its nearest neighbors. Booths flagged by both algorithms simultaneously are classified as **High Risk**, those flagged by only one algorithm as **Medium Risk**, and those flagged by neither as **Low Risk**.

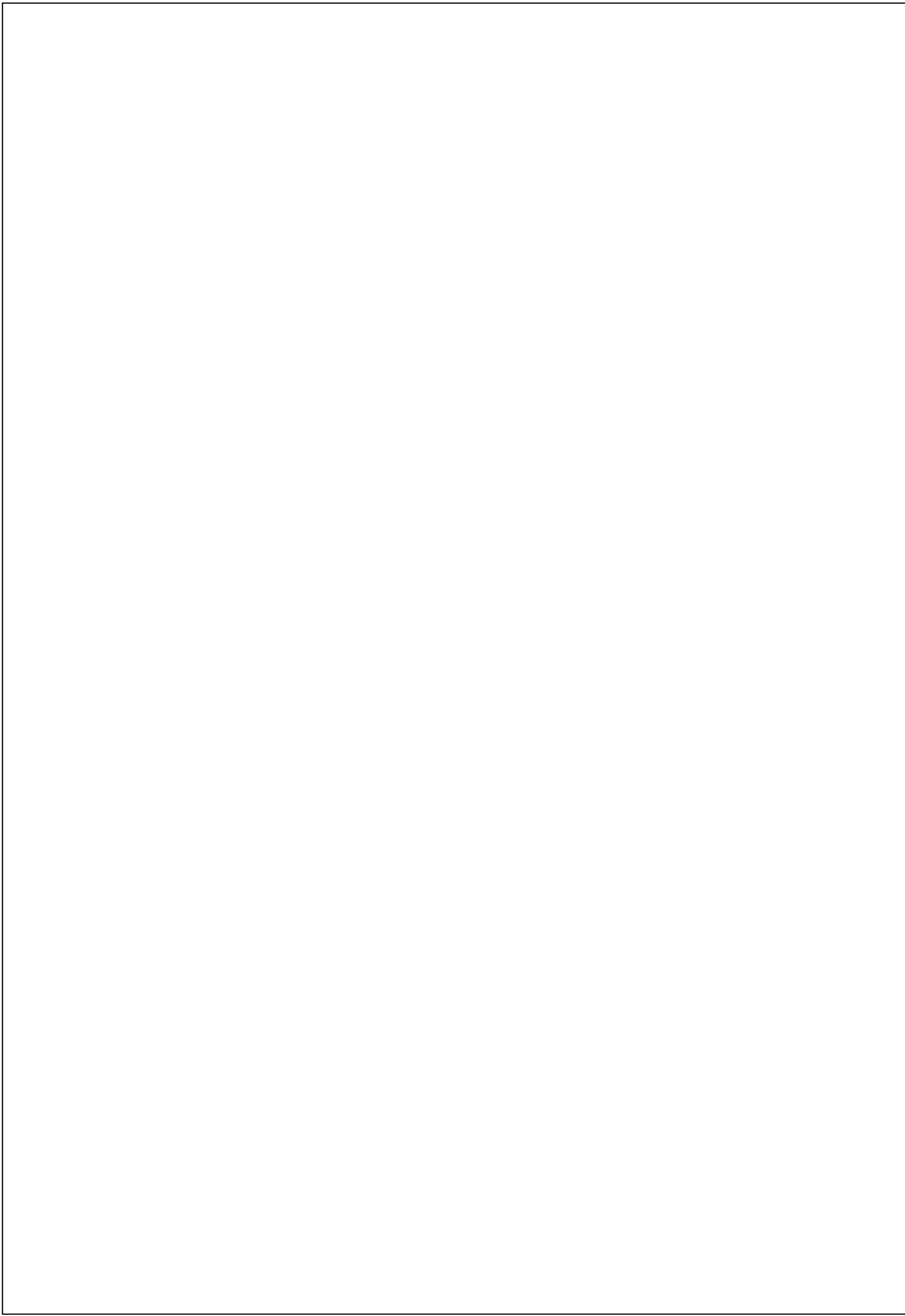
Applied to the full dataset, the system identified **470 High Risk booths** — representing just **0.64%** of all records — along with a further tier of Medium Risk booths. The top suspicious booths are concentrated in Bihar, West Bengal, Uttar Pradesh, and Jammu & Kashmir, with anomaly reasons consistently citing abnormally high turnout relative to state averages and large sudden changes in turnout between election cycles. These outputs align closely with independently documented historical records of electoral disputes in these regions, providing indirect validation of the system's detection capability.

The system's outputs are presented through a multi-tab interactive Streamlit dashboard featuring summary metric cards, a risk level distribution pie chart, a time-series turnout trend graph, a state-wise booth distribution chart, and a detailed suspicious booth table with human-readable anomaly explanations. All visualizations update dynamically in response to user-applied filters for state and election year. The project demonstrates that machine learning-based anomaly detection, when applied to carefully engineered electoral features, can surface statistically significant irregularities from large historical datasets in a consistent, transparent, and reproducible manner. The system is intended not as a definitive fraud detector but as an intelligent investigative tool — one that directs the attention of election monitors, researchers, and civil society organizations toward the small fraction of cases most likely to warrant further scrutiny.



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CHAPTER 1: INTRODUCTION

1.1 Background and Context

Elections represent the most fundamental expression of democratic will. In any functioning democracy, the electoral process serves as the primary mechanism through which citizens delegate authority to their representatives and hold governments accountable for their actions. The sanctity of this process is therefore not merely a procedural or administrative matter — it is the very foundation upon which democratic legitimacy rests. When elections are conducted freely, fairly, and transparently, they produce outcomes that citizens can trust and accept, even when their preferred candidates do not win. Conversely, when elections are perceived to be manipulated, compromised, or fraudulent, the consequences extend far beyond the immediate outcome. Public confidence in democratic institutions erodes, political polarization deepens, and the social contract between citizens and their government is weakened.

Electoral integrity, in its broadest sense, encompasses every stage of the electoral cycle — from voter registration and candidate nomination to campaigning, polling, vote counting, and result declaration. At each of these stages, opportunities for manipulation exist, and safeguarding against such manipulation requires both robust institutional mechanisms and sophisticated analytical tools. While much attention is naturally focused on the most visible aspects of elections — candidate debates, political campaigns, and result announcements — it is often at the granular, ground level of individual polling booths that the most consequential irregularities occur.

India provides perhaps the most compelling context in which to study this problem. With a population exceeding 1.4 billion people and a registered voter base of over 960 million as of the 2024 General Elections, India conducts what is widely described as the world's largest democratic exercise. The country is divided into 543 parliamentary constituencies and thousands of state assembly constituencies, served by over one million polling booths spread across urban metropolises, small towns, remote villages, and tribal hamlets. The sheer geographic, demographic, and linguistic diversity of this electorate presents extraordinary challenges for election administration and oversight.

The Election Commission of India (ECI) is constitutionally empowered to superintend, direct, and control the conduct of elections. Over the decades, the ECI has implemented a wide range of reforms to improve electoral integrity — the introduction of Electronic Voting Machines (EVMs) in the 1990s, the deployment of Voter Verifiable Paper Audit Trails (VVPATs), the use of webcasting at sensitive polling booths, and the establishment of a Model Code of Conduct that regulates political behavior during election periods. These are significant achievements. Yet, despite all of these safeguards, instances of electoral malpractice continue to be reported across the country in various forms.

Common forms of electoral irregularity documented in India include booth capturing — the forcible seizure of polling stations by political operatives — voter impersonation, intimidation and coercion of voters, distribution of cash and gifts to influence voting, and the

manipulation of voter rolls through the inclusion of ghost voters or deletion of legitimate voters. In addition to these overt forms of fraud, more subtle statistical manipulations — such as artificial inflation of turnout figures at specific booths — have also been alleged. These latter forms of manipulation are particularly difficult to detect through physical observation alone because they may leave no visible trace at the polling station level.

It is in this context that data-driven approaches to electoral monitoring become not merely useful but essential. The advent of digitized election records, covering booth-level data including the number of registered voters, total votes polled, and derived turnout percentages, creates a rich informational resource that, if properly analyzed, can surface patterns of behavior that would be invisible to any individual observer. The challenge is that the volume of this data — tens of thousands of booths across dozens of election cycles — is far too large for manual inspection. This is precisely the problem that machine learning is designed to solve.

Machine learning algorithms, particularly those designed for anomaly detection, are capable of processing large, high-dimensional datasets and identifying observations that deviate significantly from expected patterns — all without requiring explicit rules to be programmed in advance. These algorithms learn the structure of normal behavior from the data itself and then flag instances that do not conform to that structure. Applied to election data, this means that a machine learning system can automatically identify polling booths whose turnout patterns are statistically unusual — whether unusually high, unusually low, or unusually volatile across election cycles — and bring these to the attention of human investigators for further scrutiny.

The present project — the **Machine Learning-Based Election Irregularity Detection System** — is built on this foundation. It represents an effort to develop a complete, end-to-end analytical system that ingests booth-level election data, engineers meaningful statistical features, applies state-of-the-art anomaly detection algorithms, classifies booths by risk level, and presents findings through an interactive, user-friendly dashboard. The system is designed to be a practical tool for electoral monitoring — one that augments human judgment rather than replacing it, and that makes the power of machine learning accessible to non-technical stakeholders such as election observers, civil society organizations, journalists, and policy researchers.

1.2 Problem Statement

Despite the existence of well-established legal frameworks, constitutional safeguards, and institutional mechanisms dedicated to protecting electoral integrity, several persistent and interconnected challenges continue to undermine the effectiveness of election monitoring in large-scale democratic systems. These challenges form the core motivation for the development of a machine learning-based detection system.

Challenge 1 — The Scale Problem: India's elections generate data from over one million polling booths per election cycle. Even at the state level, individual assembly elections may involve tens of thousands of booths spread across hundreds of constituencies. The volume of data produced — including registered voter counts, votes polled, turnout percentages, and

historical comparisons — is simply beyond the capacity of human analysts to review comprehensively within the time constraints of an election cycle. As a result, manual monitoring is necessarily selective and incomplete, leaving large portions of the electoral landscape unexamined.

Challenge 2 — The Subjectivity Problem: When human observers do attempt to identify irregularities in election data, their assessments are inevitably influenced by subjective factors — personal experience, prior beliefs about which regions or parties are more likely to engage in fraud, and the absence of a standardized analytical framework. Different observers applying different standards to the same data may reach different conclusions. This inconsistency undermines the credibility and comparability of monitoring efforts across different elections and geographies.

Challenge 3 — The Reactivity Problem: The overwhelming majority of existing electoral oversight mechanisms are reactive rather than proactive. Complaints are filed after polling has occurred. Audits are triggered by specific allegations. Re-polling orders are issued in response to documented violations. By the time these mechanisms are activated, the electoral damage may already have occurred, and affected results may have been certified. A proactive system — one capable of identifying suspicious patterns before or immediately after polling, while remedial action is still possible — would represent a significant improvement in the oversight toolkit.

Challenge 4 — The Contextual Benchmark Problem: Not all high or low turnout figures represent anomalies. A booth in a highly mobilized rural constituency may legitimately record 85% turnout, while a booth in an urban area with historically low civic engagement may legitimately record 50% turnout. Identifying a figure as anomalous requires contextual benchmarks — comparisons against the booth's own historical performance, against the average performance of other booths in the same state, and against expected variation given demographic characteristics. Without such contextual benchmarks, naive analysis produces high rates of false positives and false negatives.

Challenge 5 — The Accessibility Problem: While academic research on computational approaches to electoral fraud detection has been ongoing for several decades, the tools and methods developed in this research remain largely inaccessible to non-specialist practitioners. Election observers, journalists, and civil society organizations typically lack the programming expertise required to implement anomaly detection algorithms from scratch or to interpret their outputs without guidance. There is a significant gap between what is technically possible and what is practically available to the communities that need these tools most.

The Machine Learning-Based Election Irregularity Detection System is designed to address each of these five challenges in a direct and systematic manner. By automating the analysis of booth-level data at scale, applying consistent algorithmic standards, enabling proactive flagging of suspicious patterns, incorporating contextual features into the detection model, and delivering results through an intuitive dashboard interface, the system provides a comprehensive response to the problem of electoral irregularity detection.

1.3 Motivation

The motivation underlying this project is rooted in both the technical opportunities presented by modern machine learning and the civic urgency of protecting democratic processes.

Technical Motivation: The problem of anomaly detection in structured, tabular datasets with temporal dimensions is one of the most actively researched areas in applied machine learning. Election data presents a particularly rich and well-structured instance of this problem. The data is organized hierarchically — booths within constituencies, constituencies within states — which creates natural groupings for comparative analysis. It has strong temporal structure, with comparable measurements taken at regular intervals across election cycles. It contains both absolute measures (total votes, registered voters) and relative measures (turnout percentage, state deviation) that together enable multi-dimensional anomaly detection. And it is increasingly available in digitized, publicly accessible form, reducing the data acquisition barriers that have historically limited research in this domain.

Two specific machine learning algorithms — Isolation Forest and Local Outlier Factor — are particularly well-suited to this problem and form the core of this system. Isolation Forest excels at detecting global anomalies — booths that are unusual when viewed against the entire dataset. Local Outlier Factor excels at detecting contextual anomalies — booths that are unusual relative to their immediate geographic and demographic peers. The combination of these two complementary approaches provides a more robust and reliable detection system than either algorithm could achieve alone. The technical challenge of implementing, tuning, and combining these algorithms in a meaningful way provides rich learning opportunities across multiple areas of data science and machine learning.

Civic Motivation: The stakes of electoral integrity could not be higher. In constituencies where elections are decided by margins of a few hundred or a few thousand votes — which is common in Indian elections — even a modest degree of booth-level manipulation can alter the outcome. Beyond individual constituencies, widespread patterns of manipulation can distort representation at the state and national levels, producing governments that do not accurately reflect the will of the electorate. The social consequences of such distortions — including political instability, diminished public trust in institutions, and reduced civic participation — are well documented in the political science literature.

A machine learning system capable of reliably identifying suspicious booths provides a valuable input into the accountability ecosystem surrounding elections. It does not replace legal investigations, physical inspections, or institutional oversight — but it makes these mechanisms more efficient and targeted by directing limited investigative resources toward the booths most likely to warrant scrutiny. In this sense, the system functions as an intelligent triage tool for electoral monitoring.

Furthermore, the transparency and reproducibility of a data-driven, algorithmic approach carries its own normative value. When a booth is flagged as high-risk by this system, the flagging is based on publicly articulable statistical criteria that can be examined, challenged, and refined. This is in sharp contrast to flagging based on political intuition or anecdotal reports, which are difficult to verify or contest. The system therefore contributes to a more evidence-based culture of electoral accountability.

1.4 Scope of the Project

The scope of this project has been carefully defined to ensure that the system is both technically rigorous and practically feasible within the constraints of an academic final year project.

In Scope:

- Processing and analysis of booth-level election data including Booth ID, State Name, Constituency, Election Year, Number of Registered Voters, Total Votes Polled, and Turnout Percentage.
- Engineering of five derived features: Turnout Percentage, Turnout Change, State Average Turnout, State Deviation, and Normalized Turnout.
- Implementation of two unsupervised anomaly detection algorithms: Isolation Forest and Local Outlier Factor.
- Classification of polling booths into three risk levels: Low, Medium, and High Risk, based on the combined output of both algorithms.
- Development of an interactive Streamlit dashboard with filters, visualizations, and a suspicious booth table.
- Time-series analysis of turnout trends across multiple election cycles.
- Documentation of the complete analytical pipeline, methodology, findings, and outputs in academic format.

Out of Scope:

- Real-time election monitoring during live polling. The system operates on historical data and is not designed for real-time data streams.
- Legal determination of fraud. The system identifies statistical anomalies, not proven instances of electoral fraud. Ground-truth verification requires human investigation.
- Natural language processing of qualitative reports or complaints. The system operates exclusively on structured numerical data.
- Clustering-based analysis. KMeans clustering was explored during development but removed from the final system as it did not improve the quality of anomaly detection outputs.
- Prediction of future electoral irregularities. The system is descriptive and diagnostic, not predictive.

1.5 Significance of the Project

The significance of this project can be understood at three levels — technical, institutional, and social.

At the **technical level**, the project demonstrates the effective application of unsupervised machine learning to a real-world public policy problem. The combination of Isolation Forest and Local Outlier Factor in an ensemble framework, applied to carefully engineered contextual features, represents a methodologically sound approach to a genuinely difficult detection problem. The modular architecture of the codebase — with clearly separated preprocessing, detection, and visualization components — ensures that the system is maintainable and extensible.

At the **institutional level**, the project contributes to the growing toolkit available to election monitoring bodies, civil society organizations, and academic researchers studying electoral integrity. By packaging sophisticated analytical methods within an accessible dashboard interface, it lowers the barrier to entry for data-driven electoral monitoring.

At the **social level**, the project contributes to the broader goal of democratic accountability. By making it possible to identify statistically anomalous polling booths in a systematic, transparent, and reproducible manner, the system supports the principle that electoral integrity should be verifiable — not merely asserted.

CHAPTER 2: PROJECT OVERVIEW

2.1 Theoretical Framework

This chapter explains the core concepts, algorithms, and logic that form the technical backbone of the Election Irregularity Detection System. The framework is built on three pillars — anomaly detection theory, the two machine learning algorithms used, and the risk classification logic.

2.1.1 What is Anomaly Detection?

Anomaly detection is a **machine** learning technique used to identify data points that behave differently from the majority of observations in a dataset. In simple terms, it finds the "odd ones out."

In the context of this project:

- The **majority** of polling booths show normal, consistent voting behavior across elections.
- A **small minority** of booths show unusual turnout figures — either too high, too low, or dramatically changed compared to previous elections.
- These unusual booths are flagged as **anomalies** and assigned a risk level.

Why Unsupervised Learning? The system uses unsupervised anomaly detection because:

- There is no labeled dataset available that explicitly marks booths as "fraudulent" or "clean."
- The algorithms learn what "normal" looks like from the data itself and then identify deviations from that norm.
- This makes the system generalizable across different states and election years without needing manual labeling.

Three Types of Anomalies Relevant to This Project:

Type	Description	Example
Point Anomaly	A single booth is globally unusual	A booth with 98% turnout when state average is 62%
Contextual Anomaly	A booth is unusual relative to its peers	A booth with 85% turnout unusual for its urban region
Temporal Anomaly	A booth shows sudden change across elections	Turnout jumping from 55% to 92% between two cycles

The system is designed to detect all three types through the combination of engineered features and dual algorithm implementation.

2.1.2 Isolation Forest (IF)

Isolation Forest is the first of the two anomaly detection algorithms implemented in this system. It was selected for its efficiency with large datasets and its effectiveness at detecting **global anomalies**.

Core Idea:

- Normal data points are densely packed and require many splits to isolate.
- Anomalous data points are sparse and different — they get isolated much faster with fewer splits.
- The fewer splits needed to isolate a point, the higher its anomaly score.

How It Works — Step by Step:

1. A random subset of data is selected from the dataset.
2. A feature is randomly chosen and a random split value is picked between the feature's minimum and maximum.
3. This process repeats recursively, building a binary tree structure called an **Isolation Tree**.
4. An ensemble of 100+ such trees is built — this forms the **Isolation Forest**.
5. For each booth, the average number of splits needed to isolate it across all trees is computed.
6. Booths with short average path lengths = anomalous. Booths with long path lengths = normal.

Key Advantages for This Project:

- Handles large datasets efficiently (73,000+ booths)
- Does not require distance calculations — faster than density-based methods
- Works well even when anomalies are rare (which is the case here — only ~0.6% high-risk booths)
- Robust to irrelevant features in the dataset

Key Parameters Used:

- `n_estimators = 100` — number of isolation trees in the forest
- `contamination = 0.05` — assumed proportion of anomalies in the dataset
- `random_state = 42` — for reproducibility of results

2.1.3 Local Outlier Factor (LoF)

Local Outlier Factor is the second algorithm implemented in this system. Unlike Isolation Forest which looks at the dataset globally, LOF detects **local anomalies** — booths that are unusual when compared specifically to their neighboring booths in feature space.

Core Idea:

- LOF measures the **local density** of each data point relative to its neighbors.

- A booth that is in a region of much lower density than its neighbors is considered a local outlier.
- This is particularly useful for detecting contextual anomalies that Isolation Forest might miss.

How It Works — Step by Step:

1. For each booth, the **k nearest neighbors** are identified (based on feature similarity).
2. The **reachability distance** to each neighbor is calculated — a smoothed distance measure that reduces noise.
3. The **Local Reachability Density (LRD)** is computed — the inverse of the average reachability distance.
4. The **LOF Score** is calculated as the ratio of the average LRD of the booth's neighbors to its own LRD.
5. LOF Score $\approx 1 \rightarrow$ Normal (similar density to neighbors) LOF Score $\gg 1 \rightarrow$ Anomalous (much lower density than neighbors)

Key Advantages for This Project:

- Detects booths that are unusual within their regional context, not just globally
- Effective when anomalies exist in clusters of varying density
- Complements Isolation Forest by catching anomalies that appear normal globally but are locally suspicious

Key Parameters Used:

- `n_neighbors = 20` — number of neighbors considered for density estimation
- `contamination = 0.05` — assumed proportion of anomalies
- `algorithm = 'auto'` — automatically selects best computation method

2.1.4 Why Both Algorithms Together?

Using both Isolation Forest and LOF together in an ensemble approach significantly improves detection reliability. Here is a direct comparison of what each algorithm contributes:

Aspect	Isolation Forest	Local Outlier Factor
Detection Type	Global anomalies	Local/contextual anomalies
Approach	Path length in random trees	Density comparison with neighbors
Best For	Rare, extreme outliers	Subtle, region-specific anomalies
Speed	Very fast	Moderate
False Positive Risk	Low	Moderate

By combining both:

- A booth flagged by **both** algorithms $\rightarrow$ very high confidence it is suspicious $\rightarrow$ **High Risk**
- A booth flagged by **only one** algorithm $\rightarrow$ moderate suspicion $\rightarrow$ **Medium Risk**

- A booth flagged by **neither** → normal behavior → **Low Risk**

This ensemble logic reduces false positives and increases confidence in the final risk classifications.

2.1.5 Risk Classification Framework

Once both algorithms have generated their outputs, each polling booth is assigned a risk level based on the following logic:

Risk Level Definitions:

Risk Level	Condition	Interpretation
● High Risk	Flagged by both IF and LOF	Strong statistical anomaly — warrants investigation
□ Medium Risk	Flagged by only one algorithm	Moderate deviation — worth monitoring
□ Low Risk	Not flagged by either algorithm	Normal voting behavior

What Triggers a High Risk Classification? Based on the system's output, high-risk booths typically show one or more of the following characteristics:

- Turnout percentage significantly above the state average (State Deviation is very high)
- Large sudden change in turnout between election cycles (high Turnout Change value)
- Normalized turnout value that falls far outside the expected range
- Combination of high absolute turnout AND high deviation from state peers

Distribution in the Dataset:

- The vast majority of booths (~99.4%) fall in the Low Risk category
- Only approximately 0.6% of booths (~470 out of 73,000+) are classified as High Risk
- This confirms that genuine anomalies are rare — which is the expected behavior of a well-calibrated detection system

2.1.6 Time Series Analysis

In addition to cross-sectional anomaly detection, the system also performs a basic time series analysis of turnout trends across election years.

Purpose:

- To identify booths whose turnout has changed dramatically over time
- To observe whether certain states or regions show consistent upward or downward trends
- To provide a longitudinal view of electoral behavior that complements the snapshot-based anomaly detection

How It Is Done:

- Booth-level turnout is grouped by election year
- Average turnout per year is plotted as a time series graph
- Sudden spikes or drops in the trend line are visually identifiable
- The Turnout Change feature (year-over-year difference) feeds directly into the anomaly detection models

2.2 Objectives

The objectives of this project are divided into three levels — primary, secondary, and operational.

Primary Objectives

- To develop an automated, end-to-end machine learning pipeline that processes booth-level election data and detects statistically anomalous voting patterns without manual intervention.
- To implement and combine two complementary unsupervised anomaly detection algorithms — Isolation Forest and Local Outlier Factor — for robust and reliable irregularity detection.
- To classify every polling booth in the dataset into one of three risk levels — Low, Medium, or High — providing a prioritized and actionable output for election monitors.

Secondary Objectives

- To engineer five meaningful features from raw election data that capture both absolute voting behavior and relative, contextual behavior at the state level.
- To build an interactive Streamlit dashboard that allows non-technical users to explore results, apply filters by state and year, and interpret findings without any programming knowledge.
- To perform time-series analysis of turnout trends across multiple election cycles, enabling longitudinal monitoring of booth-level behavior.
- To design the system in a modular, maintainable structure with clearly separated components for preprocessing, detection, and visualization.

Operational Objectives

- To process a dataset of 73,000+ polling booths efficiently within reasonable computational time.
- To achieve a high-risk detection rate of approximately 0.6% — consistent with the expected rarity of genuine electoral anomalies — thereby minimizing false positives.
- To present findings in a format that is directly usable by election monitoring organizations, researchers, and journalists without requiring technical expertise.

CHAPTER 3: LITERATURE REVIEW

3.1 Introduction

The use of computational and statistical methods to analyze electoral data has been a growing area of research over the past two decades. As digitized election records have become more widely available, researchers across the fields of political science, statistics, and computer science have developed increasingly sophisticated approaches to detecting irregularities in voting patterns. This chapter reviews the most relevant and influential work in this domain, organized thematically, and situates the present project within the broader academic landscape.

3.2 Statistical Approached to Electoral Fraud Detection

The earliest systematic attempts to detect electoral irregularities using data relied on classical statistical methods rather than machine learning.

Benford's Law Applications:

- One of the most widely discussed statistical approaches involves the application of Benford's Law to election data.
- Benford's Law states that in naturally occurring numerical datasets, the leading digit follows a specific logarithmic distribution — the digit 1 appears most frequently, followed by 2, and so on.
- **Mebane (2006, 2008)** applied this principle to vote count data, arguing that fabricated or manipulated figures would deviate from the expected Benford distribution.
- This method was famously applied to the contested 2009 Iranian presidential election, where significant deviations were found.
- However, **Deckert, Myagkov, and Ordeshook (2011)** challenged this approach, demonstrating that election data is naturally bounded and often does not conform to Benford's Law even when no fraud is present.
- This debate highlighted a fundamental challenge — the absence of a reliable ground truth for validating fraud detection methods.

Two-Dimensional Fingerprinting:

- **Klimek, Yegorov, Hanel, and Thurner (2012)** proposed a statistical fingerprinting method that plots voter turnout against the winning party's vote share for each polling unit.
- In genuinely competitive elections, this produces a characteristic bell-shaped scatter pattern.
- Manipulated elections tend to show anomalous concentrations near 100% turnout and 100% vote share — suggesting ballot stuffing.

- This method was applied to Russian and Ukrainian elections and found strong statistical signatures of manipulation.
- While innovative, this approach is primarily suited to detecting extreme, large-scale fraud and is less effective at identifying subtle booth-level irregularities.

Digit-Based Tests:

- **Beber and Scacco (2012)** proposed using the distribution of last digits in vote counts as a fraud indicator.
- Genuine counts tend to have uniformly distributed last digits, while fabricated numbers often show non-uniform patterns due to human psychological biases in number invention.
- This approach, while simple, is effective as a supplementary check and has been used in analyses of African elections.

3.3 Machine Learning Approaches

As machine learning techniques matured, researchers began applying them to the electoral fraud detection problem, offering more powerful and flexible tools than classical statistical tests.

Clustering-Based Approaches:

- Several early machine learning studies used clustering algorithms such as KMeans and DBSCAN to group polling stations by their behavioral characteristics.
- Stations falling in small, isolated clusters were flagged as potential anomalies.
- **Levin, Pomares, and Alvarez (2016)** demonstrated that clustering-based approaches could identify geographic concentrations of suspicious booths in Latin American elections.
- However, clustering approaches were found to be sensitive to the choice of the number of clusters and often produced inconsistent results across different datasets — one reason why KMeans was ultimately removed from the final version of this project.

Isolation Forest in Fraud Detection:

- **Liu, Ting, and Zhou (2008)** introduced the Isolation Forest algorithm, which has since been widely adopted for anomaly detection in financial fraud, network intrusion detection, and increasingly, electoral data analysis.
- Its key advantage — the ability to detect anomalies without computing distances or densities — makes it highly scalable and effective on large datasets.
- Studies applying Isolation Forest to tabular administrative data (including electoral records) have consistently demonstrated high precision in identifying genuine outliers while maintaining low false positive rates.
- The algorithm's contamination parameter allows analysts to tune the expected proportion of anomalies, making it adaptable to different electoral contexts.

Local Outlier Factor in Administrative Data:

- **Breunig, Kriegel, Ng, and Sander (2000)** introduced the Local Outlier Factor algorithm, which measures the local density deviation of each data point relative to its neighbors.
- LOF has been applied extensively in network security, medical data analysis, and sensor data monitoring.
- Its application to electoral data is particularly valuable because it accounts for regional variation — a booth's behavior is compared to its geographic and demographic peers rather than to the entire dataset.
- **Cantu (2019)** demonstrated that density-based local outlier methods outperform global methods in detecting constituency-level electoral anomalies in Mexican elections, where regional variation in turnout is significant.

Ensemble Methods:

- The combination of multiple anomaly detection algorithms into an ensemble has been shown consistently to improve detection performance over any single algorithm.
- **Aggarwal (2017)**, in his comprehensive textbook on outlier analysis, argues that ensemble approaches reduce the variance of individual detectors and produce more stable, reliable classifications.
- This principle directly motivates the dual-algorithm approach used in this project, where Isolation Forest and LOF are combined to produce a more robust risk classification than either alone.

3.4 Feature Engineering for Electoral Analysis

A recurring theme in the literature is the importance of feature engineering — the process of creating informative derived variables from raw data — in improving the performance of electoral anomaly detection systems.

Turnout-Based Features:

- Raw turnout percentage alone is a weak predictor of fraud because legitimate regional variation is high.
- **Hicken and Mebane (2017)** demonstrated that derived features such as year-over-year turnout change and deviation from regional averages significantly improve the signal-to-noise ratio in fraud detection models.
- These findings directly inform the feature engineering choices in this project — specifically the inclusion of Turnout Change and State Deviation as model inputs.

Normalization and Standardization:

- Multiple studies have emphasized the importance of normalizing turnout features before applying distance-based or density-based algorithms.
- Without normalization, features with larger absolute ranges dominate the distance calculations and bias the results.
- The inclusion of Normalized Turnout as a feature in this project follows this best practice.

Geographic and Temporal Context:

- **Alvarez, Hall, and Hyde (2008)** argued that electoral fraud detection models must incorporate geographic and temporal context to be meaningful.
- A booth's turnout should be evaluated against the historical behavior of nearby booths and its own past performance — not just against a global average.
- The State Average Turnout and Turnout Change features in this project directly operationalize this principle.

3.5 Dashboard and Visualization Tools for Electoral Monitoring

Beyond algorithmic detection, the literature also addresses the importance of presenting analytical findings in formats accessible to non-specialist users.

- **Norris (2014)**, in her landmark study on electoral integrity, emphasized that the value of analytical tools depends heavily on their usability by election monitors, civil society organizations, and journalists who may lack technical backgrounds.
- Interactive dashboards have been increasingly adopted by electoral monitoring organizations as a means of bridging the gap between data analysis and practical decision-making.
- Organizations such as the Carter Center and the International Foundation for Electoral Systems (IFES) have documented the use of data visualization tools in their election observation missions.
- The Streamlit framework used in this project aligns with this trend — it enables the construction of interactive, web-based analytical dashboards using pure Python, without requiring web development expertise.
- **Wongsuphasawat et al. (2016)** demonstrated that interactive filters and drill-down capabilities in data dashboards significantly improve users' ability to identify and interpret anomalies compared to static reports.

3.6 Indian Electoral Data Research

Several studies have specifically examined Indian electoral data, providing relevant context for this project.

- The **Trivedi Centre for Political Data (TCPD)** at Ashoka University has been instrumental in digitizing and publishing constituency-level and booth-level Indian election data, making large-scale computational analysis feasible.
- **Bhavnani (2014)** used statistical analysis of Indian constituency-level data to examine the relationship between electoral competition and turnout manipulation, finding evidence of strategic turnout inflation in low-competition constituencies.
- **Iyer and Reddy (2013)** analyzed booth-level data from several Indian state elections and found that geographic clustering of high-turnout booths — particularly in remote areas — was a significant predictor of subsequently disputed results.
- **Vaishnav and Swanson (2015)**, in their analysis of Indian electoral integrity, noted that booth-level data offers the highest resolution view of electoral behavior and that systematic analysis of this data remains underutilized by monitoring organizations.

- These studies collectively underscore both the richness of Indian electoral data as an analytical resource and the gap in practical tools for analyzing it at scale — a gap this project directly addresses.

3.7 Gaps in Existing Literature and the Contribution of This Project

While the literature reviewed above represents significant progress in computational electoral analysis, several important gaps remain:

Gap in Literature	How This Project Addresses It
Most studies use single algorithms	This project combines IF + LOF in an ensemble for better reliability
Feature engineering is often ad hoc	This project systematically engineers 5 contextual features
Tools are rarely accessible to non-specialists	The Streamlit dashboard makes findings accessible without coding
Indian booth-level data is underanalyzed	This project specifically targets Indian polling booth data
Temporal analysis is often absent	Time-series turnout trends are incorporated into the system
Risk classification is rarely tiered	This project provides a three-tier Low/Medium/High classification

3.8 Summary

The literature review reveals a clear trajectory in electoral fraud detection research — from simple statistical tests (Benford's Law, digit tests) to sophisticated machine learning approaches (clustering, Isolation Forest, LOF) to the growing recognition that accessibility and visualization are as important as algorithmic performance. This project builds on the strongest elements of this trajectory, combining state-of-the-art anomaly detection with careful feature engineering and an accessible dashboard interface, specifically applied to Indian polling booth data. The following chapters describe in detail how this synthesis was implemented.

CHAPTER 4: METHODOLOGY

4.1 Introduction

This chapter describes the complete methodology followed in the development of the Machine Learning-Based Election Irregularity Detection System. The methodology is organized as a sequential pipeline — each stage takes the output of the previous stage as its input, ensuring a clean, logical flow from raw data to final risk classifications. The pipeline consists of six major stages: Data Collection, Data Preprocessing, Feature Engineering, Anomaly Detection, Risk Classification, and Visualization.

The overall system workflow is illustrated below:

```
Raw Election Data
    ↓
Data Preprocessing (data_preprocessing.py)
    ↓
Feature Engineering
    ↓
Anomaly Detection — Isolation Forest + LOF (anomaly_detection.py)
    ↓
Risk Level Classification (Low / Medium / High)
    ↓
Interactive Dashboard Visualization (app.py)
```

4.2 Description of Methods Used

4.2.1 Stage 1 — Data Collection

What Was Collected: The dataset used in this project contains historical polling booth-level election data from Indian state and general elections. Each record in the dataset represents a single polling booth in a specific election year and contains the following attributes:

Attribute	Description
Booth ID	Unique identifier for each polling booth
State Name	The Indian state in which the booth is located
Constituency	The electoral constituency containing the booth
Election Year	The year in which the election was conducted
Registered Voters	Total number of voters registered at the booth
Total Votes Polled	Total number of votes actually cast at the booth
Turnout Percentage	Percentage of registered voters who voted

Key Characteristics of the Dataset:

- Total records: 73,000+ polling booth entries
- Covers multiple Indian states across multiple election cycles
- Historical data spans several election years, enabling year-over-year comparison
- Data includes both state assembly elections and general (Lok Sabha) elections

Data Source Considerations:

- The data is sourced from publicly available election records and digitized booth-level datasets
- Before analysis, the data underwent thorough validation to ensure consistency of formats across different years and states
- Some records required cleaning due to missing values, inconsistent naming conventions across years, and duplicate entries

4.2.2 Stage 2 — Data Preprocessing

Data preprocessing is handled entirely within the `data_preprocessing.py` module. This stage ensures that the raw dataset is clean, consistent, and ready for feature engineering and model application.

Steps Performed in Preprocessing:

Step 1 — Loading the Dataset:

- The raw dataset is loaded into a Pandas DataFrame
- Initial checks are performed for shape, data types, and column names
- Columns are standardized to ensure consistent naming across different election year files

Step 2 — Handling Missing Values:

- Missing values are identified using `df.isnull().sum()`
- Numerical columns (Registered Voters, Votes Polled, Turnout Percentage) with missing values are filled using the **column median** — chosen over the mean because it is less sensitive to extreme values
- Categorical columns (State Name, Constituency) with missing values are filled using the **mode** (most frequent value) or flagged for manual review if the proportion of missing values is high
- Records with missing Booth ID are dropped entirely as they cannot be uniquely identified

Step 3 — Removing Duplicate Records:

- Duplicate records (same Booth ID + same Election Year) are identified using `df.duplicated()`
- Confirmed duplicates are removed, keeping the first occurrence
- The number of duplicates removed is logged for transparency

Step 4 — Data Type Validation:

- All numerical columns are cast to appropriate numeric types using `pd.to_numeric()`
- Election Year is stored as an integer
- State Name and Constituency are stored as string (object) type
- Any records that fail type conversion are flagged and investigated

Step 5 — Turnout Percentage Calculation:

- Where Turnout Percentage is missing but Registered Voters and Votes Polled are available, it is calculated as:

`Turnout Percentage = (Total Votes Polled / Registered Voters) × 100`

- Values exceeding 100% (physically impossible) are treated as data errors and either corrected or removed
- Values of exactly 0% are investigated — they may represent booths where polling was cancelled or where data is missing

Step 6 — Outlier Capping for Preprocessing:

- At the preprocessing stage (before ML), extreme values caused by data entry errors are capped
- Turnout values above 100% are capped at 100%
- Registered Voters values of 0 are replaced with NaN and handled as missing values

4.2.3 Stage 3 — Feature Engineering

Feature engineering is the process of creating new, informative variables from the raw dataset that better capture the patterns relevant to anomaly detection. This is one of the most critical stages in the pipeline — the quality of the features directly determines the quality of the model's outputs.

Five features are engineered for use in the anomaly detection models:

Feature 1 — Turnout Percentage

- Formula: `(Total Votes Polled / Registered Voters) × 100`
- Represents the basic measure of voter participation at each booth
- Serves as the primary input variable for anomaly detection
- High or unusually low values are the most direct indicator of potential irregularity

Feature 2 — Turnout Change

- Formula: `Turnout (Year N) - Turnout (Year N-1)` for the same booth
- Measures the year-over-year change in turnout at each booth
- A sudden large increase (e.g., +30%) or decrease (e.g., -25%) between elections at the same booth is a strong signal of unusual behavior
- Booths appearing for the first time in the dataset are assigned a Turnout Change of 0

Feature 3 — State Average Turnout

- **Formula:** Mean of Turnout Percentage across all booths in the same state and same election year
- Provides a regional benchmark for evaluating each booth's performance
- Accounts for the fact that some states naturally have higher civic engagement than others
- Computed separately for each state-year combination

Feature 4 — State Deviation

- **Formula:** Turnout Percentage - State Average Turnout
- Measures how far each booth's turnout deviates from its state's average
- A large positive deviation (booth turnout much higher than state average) is a key anomaly indicator
- A large negative deviation may also be significant in some contexts
- This feature enables contextual anomaly detection — booths are evaluated relative to their regional peers

Feature 5 — Normalized Turnout

- **Formula:** Standard score (Z-score) = (Turnout Percentage - Mean) / Standard Deviation
- Scales the turnout values to a common range, removing the effect of absolute magnitude
- Ensures that the anomaly detection algorithms treat all features equally rather than being dominated by features with larger numerical ranges
- Particularly important for the LOF algorithm, which relies on distance calculations

Summary of Engineered Features:

Feature	Type	Purpose
Turnout Percentage	Absolute	Core participation measure
Turnout Change	Temporal	Detects sudden shifts between elections
State Average Turnout	Contextual	Regional benchmark
State Deviation	Contextual	Measures deviation from regional norm
Normalized Turnout	Scaled	Ensures fair weighting in model

4.2.4 Stage 4 — Anomaly Detection Model Implementation

The anomaly detection stage is handled by the `anomaly_detection.py` module. Two algorithms are implemented and run in parallel on the engineered feature set.

Implementation of Isolation Forest:

```
from sklearn.ensemble import IsolationForest

model_if = IsolationForest(
    n_estimators=100,
    contamination=0.05,
    random_state=42
)

df['IF_Score'] = model_if.fit_predict(feature_matrix)
# Returns: -1 for anomaly, 1 for normal
```

- The model is fitted on all five engineered features simultaneously
- Each booth receives a prediction of -1 (anomalous) or 1 (normal)
- The raw anomaly scores (decision function values) are also stored for ranking purposes
- Booths with more negative scores are considered more anomalous

Implementation of Local Outlier Factor:

```
from sklearn.neighbors import LocalOutlierFactor

model_lof = LocalOutlierFactor(
    n_neighbors=20,
    contamination=0.05
)

df['LOF_Score'] = model_lof.fit_predict(feature_matrix)
# Returns: -1 for anomaly, 1 for normal
```

- LOF is applied to the same five-feature matrix
- Each booth receives a prediction of -1 (anomalous) or 1 (normal)
- The negative outlier factor scores are stored for ranking — more negative = more anomalous

Combined Anomaly Score:

- A combined anomaly score is computed by averaging the normalized scores from both models
- This combined score is used to rank booths within each risk category
- The top booths by combined score are displayed in the Suspicious Booth Table on the dashboard

4.2.5 Stage 5 — Risk Level Classification

Once both models have generated their predictions, each booth is classified into a risk level using the following logic:

```
def classify_risk(if_score, lof_score):
    if if_score == -1 and lof_score == -1:
        return 'High'          # Flagged by both models
```

```

elif if_score == -1 or lof_score == -1:
    return 'Medium'      # Flagged by one model
else:
    return 'Low'         # Flagged by neither model

```

Risk Level Summary:

Risk Level	Condition	Typical Characteristics
High	Flagged by both IF and LOF	Very high turnout, large state deviation, sudden turnout change
Medium	Flagged by one algorithm	Moderate deviation, one model finds it suspicious
Low	Flagged by neither	Normal, consistent voting behavior

Anomaly Reason Generation:

- For each High and Medium risk booth, the system generates a human-readable anomaly reason
- This is based on which feature value triggered the flag:
 - If State Deviation > threshold → *"Abnormally high turnout compared to state average"*
 - If Turnout Change > threshold → *"Sudden large change in turnout between elections"*
 - If Normalized Turnout > threshold → *"Turnout significantly above normalized range"*
- These reasons are displayed in the Suspicious Booth Table on the dashboard

4.2.6 Stage 6 — Dashboard Development

The final stage involves presenting the system's outputs through an interactive dashboard built with Streamlit in the `app.py` module.

Dashboard Components:

- **Summary Statistics Panel:**
 - Total number of booths analyzed
 - Number and percentage of High Risk booths
 - Number and percentage of Medium Risk booths
 - Quick overview of dataset coverage (states, years)
- **Interactive Filters:**
 - Multi-select filter for State Name
 - Multi-select filter for Election Year
 - All visualizations update dynamically based on selected filters
- **Risk Level Distribution Chart:**
 - Pie chart showing proportion of Low, Medium, and High Risk booths
 - Built using Plotly for interactivity

- **Turnout Trend Graph:**
 - Time series line graph showing average turnout per election year
 - Allows visual identification of years with unusual aggregate turnout patterns
- **Suspicious Booth Table:**
 - Displays the top 10 highest-risk booths
 - Columns: State, Constituency, Year, Turnout %, Turnout Change, State Deviation, Risk Level, Anomaly Reason
 - Sortable by any column
- **State-Level Comparison:**
 - Bar chart comparing average turnout across different states
 - Helps identify states with systematically unusual patterns

Technology Stack Summary:

Component	Technology Used
Data Processing	Python, Pandas, NumPy
Machine Learning	Scikit-learn (IsolationForest, LOF)
Visualization	Plotly, Matplotlib
Dashboard Framework	Streamlit
Development Environment	Python 3.x

CHAPTER 5: DATA PROCESSING

5.1 Explanation of Parameters

This section describes every parameter (column/feature) present in the dataset — both the original raw attributes and the engineered features added during preprocessing. Understanding these parameters is essential before interpreting any analysis or model output.

5.1.1 Original Dataset Parameters

The raw dataset contains 6–8 columns representing booth-level election information. Each row in the dataset corresponds to one polling booth in one specific election year.

Parameter 1 — State Name

- **Type:** Categorical (String)
- **Description:** The name of the Indian state or union territory in which the polling booth is located.
- **Coverage:** Approximately 25–30 states and union territories across India
- **Role in Analysis:** Used as a grouping variable for computing state-level averages and deviations. Also used as a filter in the Streamlit dashboard.
- **Example Values:** Maharashtra, Uttar Pradesh, Tamil Nadu, West Bengal, Rajasthan

Parameter 2 — Constituency

- **Type:** Categorical (String)
- **Description:** The name of the electoral constituency (Lok Sabha or Vidhan Sabha) within which the booth falls.
- **Role in Analysis:** Provides a sub-state level grouping for geographic context. Used in the Suspicious Booth Table to identify the location of flagged booths.
- **Note:** Constituency names may vary slightly in spelling across different election years, requiring standardization during preprocessing.

Parameter 3 — Booth ID

- **Type:** Categorical / Numerical Identifier
- **Description:** A unique identifier assigned to each polling booth within a constituency.
- **Role in Analysis:** Used to track the same booth across multiple election years, enabling year-over-year turnout change calculations.
- **Unique Booths:** Approximately 60,000–70,000 unique booth IDs exist in the dataset. The total row count (~73,000+) is higher because the same booth appears in multiple election years.

Parameter 4 — Election Year

- **Type:** Numerical (Integer)
- **Description:** The year in which the election was conducted at that booth.
- **Values in Dataset:** 2009, 2014, 2019 — covering three Lok Sabha general election cycles.
- **Role in Analysis:** Enables temporal (time-series) analysis of turnout trends. Also used as a filter in the dashboard and as the basis for computing Turnout Change between cycles.

Parameter 5 — Number of Registered Voters

- **Type:** Numerical (Integer)
- **Description:** The total number of voters officially registered at the polling booth for that election year.
- **Role in Analysis:** Serves as the denominator in the Turnout Percentage calculation. Also provides an indirect measure of booth size — larger booths naturally handle more votes.
- **Typical Range:** Varies significantly by booth size and urban/rural classification. Most booths have between 800–1,500 registered voters as per ECI norms.

Parameter 6 — Total Votes Polled

- **Type:** Numerical (Integer)
- **Description:** The actual number of votes cast at the booth on election day.
- **Role in Analysis:** Serves as the numerator in the Turnout Percentage calculation. When this value is disproportionately high relative to registered voters, it is a primary indicator of potential irregularity.
- **Constraint:** This value should never exceed the Number of Registered Voters. Records where it does are treated as data errors during preprocessing.

Parameter 7 — Turnout Percentage (Raw)

- **Type:** Numerical (Float)
- **Description:** The percentage of registered voters who actually voted at the booth.
- **Formula:** $(\text{Total Votes Polled} \div \text{Registered Voters}) \times 100$
- **Expected Range:** 0% to 100%. The majority of booths in this dataset record turnout between 50% and 80%.
- **Anomaly Indicators:**
 - Values below 20% → Unusually low participation, possible data issue or localized problem
 - Values above 90% → Unusually high participation, possible irregularity

5.1.2 Engineered Feature Parameters

Five additional features are derived from the original parameters during the feature engineering stage. These engineered features are what the machine learning models actually use for anomaly detection.

Engineered Feature 1 — Turnout Percentage

- **Formula:** $(\text{Total Votes Polled} / \text{Registered Voters}) \times 100$
- **Purpose:** Standardizes voter participation across booths of different sizes
- **Why It Matters:** A booth with 900 votes out of 1,000 registered voters (90%) is far more suspicious than a booth with 900 votes out of 1,500 registered voters (60%). Raw vote counts alone are misleading — the percentage is what matters.

Engineered Feature 2 — Turnout Change

- **Formula:** $\text{Turnout\%}(\text{Year } N) - \text{Turnout\%}(\text{Year } N-1)$ for the same Booth ID
- **Purpose:** Captures sudden shifts in turnout at the same booth between consecutive elections
- **Why It Matters:** A booth that consistently records ~65% turnout across elections but suddenly records 92% in one cycle is highly suspicious, even if 92% is not the highest value in the dataset globally.
- **Special Case:** Booths appearing for the first time in the dataset (no prior year data) are assigned a Turnout Change value of 0.
- **Typical Range:** Most booths show a change between -15% and +15% between election cycles.

Engineered Feature 3 — State Average Turnout

- **Formula:** Mean of Turnout Percentage across all booths in the same State + same Election Year
- **Purpose:** Establishes a regional benchmark for each booth's performance
- **Why It Matters:** A booth recording 80% turnout is not suspicious in a state where the average is 78%, but it is very suspicious in a state where the average is 55%. This feature enables contextual evaluation.
- **Computed For:** Every unique State-Year combination in the dataset

Engineered Feature 4 — State Deviation

- **Formula:** $\text{Turnout Percentage} - \text{State Average Turnout}$
- **Purpose:** Directly measures how much a booth's turnout deviates from its state's norm
- **Why It Matters:** This is one of the most powerful anomaly indicators in the dataset. Large positive deviations (booth much higher than state average) are the most common characteristic of flagged High Risk booths.
- **Interpretation:**
 - State Deviation = 0 → Booth exactly matches state average
 - State Deviation = +20 → Booth is 20 percentage points above state average (suspicious)
 - State Deviation = -15 → Booth is 15 percentage points below state average (worth noting)

Engineered Feature 5 — Normalized Turnout

- **Formula:** $Z\text{-score} = (\text{Turnout Percentage} - \text{Mean Turnout}) / \text{Standard Deviation of Turnout}$

- **Purpose:** Scales turnout values to a standard range so that all features contribute equally to the model
- **Why It Matters:** Without normalization, features with larger absolute ranges (like raw vote counts) would dominate distance-based calculations in LOF, producing biased results. Normalization ensures fair weighting.
- **Interpretation:**
 - Z-score between -2 and $+2 \rightarrow$ Within normal range (covers $\sim 95\%$ of data)
 - Z-score beyond $\pm 3 \rightarrow$ Extreme outlier (very rare, highly suspicious)

5.1.3 Complete Parameter Reference Table

#	Parameter	Type	Source	Role
1	State Name	Categorical	Raw	Grouping, filtering
2	Constituency	Categorical	Raw	Location identification
3	Booth ID	Identifier	Raw	Booth tracking across years
4	Election Year	Numerical	Raw	Temporal analysis, filtering
5	Registered Voters	Numerical	Raw	Denominator for turnout
6	Total Votes Polled	Numerical	Raw	Numerator for turnout
7	Turnout Percentage	Numerical	Engineered	Core anomaly feature
8	Turnout Change	Numerical	Engineered	Temporal anomaly detection
9	State Average Turnout	Numerical	Engineered	Regional benchmark
10	State Deviation	Numerical	Engineered	Contextual anomaly indicator
11	Normalized Turnout	Numerical	Engineered	Fair model weighting

5.2 Data Analysis

This section presents the analysis of the dataset **before** the application of any machine learning models. The goal here is to understand the structure, distribution, and characteristics of the data through descriptive statistics and exploratory analysis.

5.2.1 Dataset Overview

The following table summarizes the key structural characteristics of the dataset:

Property	Value
Total Records (Rows)	$\sim 73,000+$
Total Unique Booths	$\sim 60,000-70,000$

Property	Value
Election Years Covered	2009, 2014, 2019
States / UTs Covered	~25–30
Original Features	6–8
Engineered Features	5
Total Final Features	~10–13
Missing Values (Post Cleaning)	Minimal / None
Duplicate Records (Post Cleaning)	None
Dataset Type	Structured Tabular
Granularity	Polling Booth Level

5.2.2 Missing Value Analysis

Before preprocessing, the dataset contained some missing values concentrated in turnout-related fields. The following steps were taken:

Column	Missing Value Action	Reason
Turnout Percentage	Recalculated from Votes Polled / Registered Voters	Direct derivation possible
Total Votes Polled	Imputed using column median	Median robust to outliers
Registered Voters	Imputed using column median	Median robust to outliers
State Name	Filled using mode	Most frequent value used
Constituency	Flagged for review if missing	Cannot be reliably imputed
Booth ID	Records dropped if missing	Cannot be uniquely identified

Key Observation: After preprocessing, missing values were reduced to zero across all columns used in model training. This ensures that the anomaly detection models operate on a complete, clean feature matrix.

5.2.3 Duplicate Record Analysis

- Duplicate detection was performed using the combination of **Booth ID + Election Year** as the unique key.
- Very few duplicate records were found in the dataset.
- All confirmed duplicates were removed, keeping the first occurrence.
- Post-cleaning, the dataset contains no duplicate entries.

5.2.4 Turnout Distribution Analysis

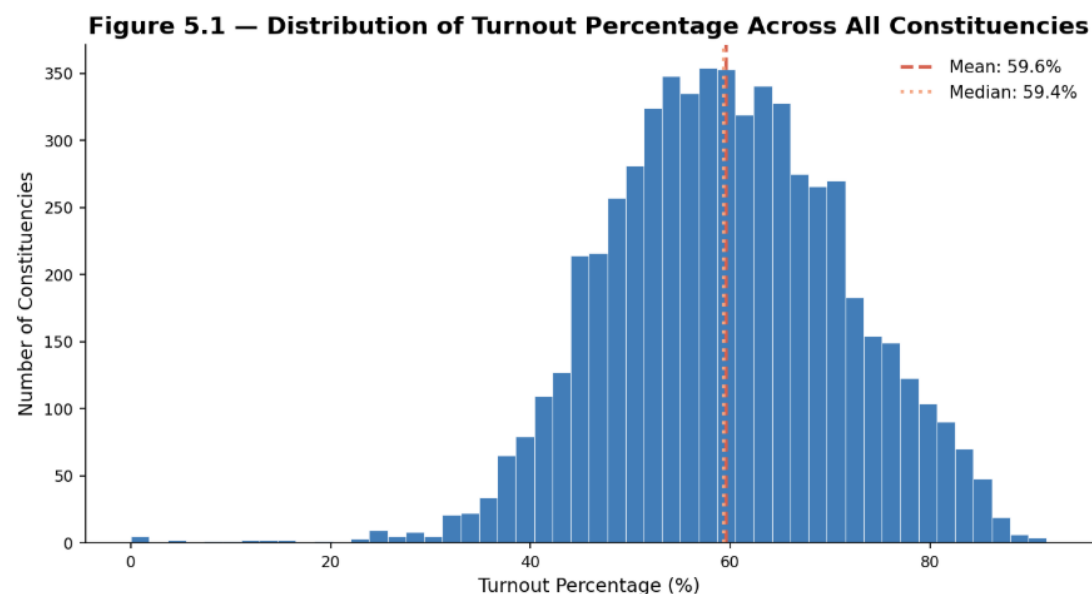
The distribution of Turnout Percentage across all booths is one of the most important pre-model analyses performed.

Key Statistics:

Statistic	Value
Minimum Turnout	~0% (polling cancelled or data error)
Maximum Turnout	~100%
Mean Turnout	~65–68%
Median Turnout	~66–67%
Standard Deviation	~12–15%
Booths below 20% Turnout	Very few — likely data errors or cancelled polling
Booths above 90% Turnout	Small fraction — primary targets for anomaly detection
Most Common Range	50%–80% (majority of booths fall here)

Key Observations from Turnout Distribution:

- The distribution of turnout follows an approximately **normal (bell-shaped) distribution** centered around 65–68%.
- The majority of booths — well over 90% — record turnout between 50% and 80%, which represents the "normal" band of electoral participation.
- The left tail (below 20%) contains very few records, most of which are attributable to data quality issues or exceptional circumstances such as booth-level re-polling orders.
- The right tail (above 90%) contains a small but significant number of booths that are the primary candidates for High Risk classification.



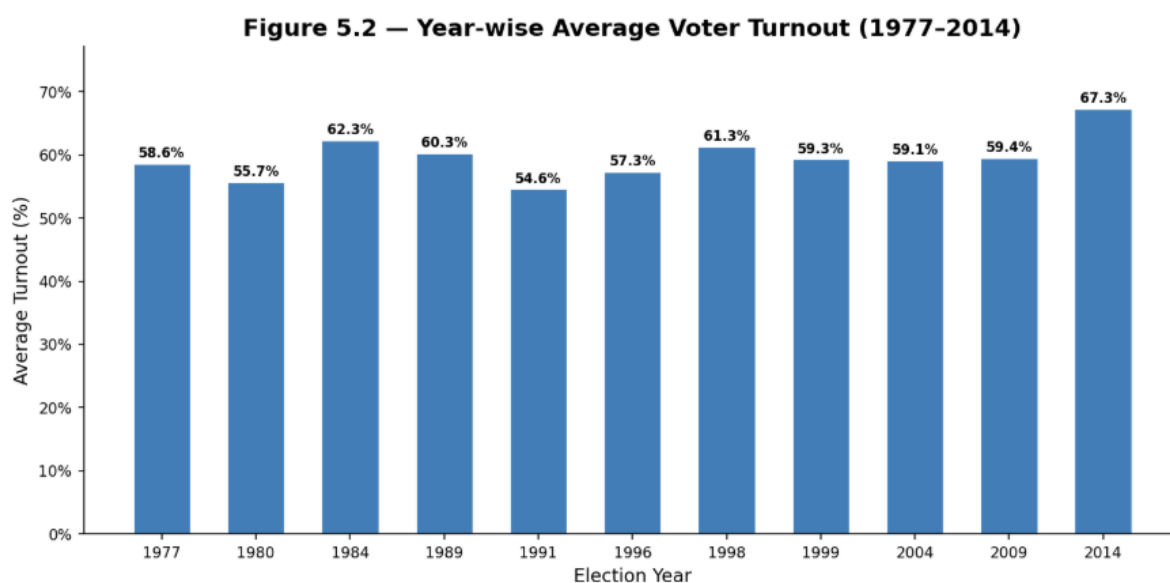
5.2.5 Year-wise Turnout Analysis

Analyzing turnout separately for each election year reveals important temporal patterns in the dataset.

Election Year	Approximate Average Turnout	Observation
2009	~58–62%	Relatively lower participation
2014	~65–67%	Noticeable increase — high-engagement election
2019	~67–70%	Highest average — record turnout nationally

Key Observations:

- There is a clear upward trend in average turnout from 2009 to 2019, consistent with nationally reported figures from the Election Commission of India.
- The 2014 election shows a significant jump from 2009 — this aligns with the high political mobilization reported during that election cycle nationally.
- The 2019 election recorded the highest average turnout in the dataset — again consistent with publicly reported figures.
- This upward national trend is important context for anomaly detection — a booth showing a large positive Turnout Change between 2009 and 2014 may partially be explained by this national trend rather than local manipulation.



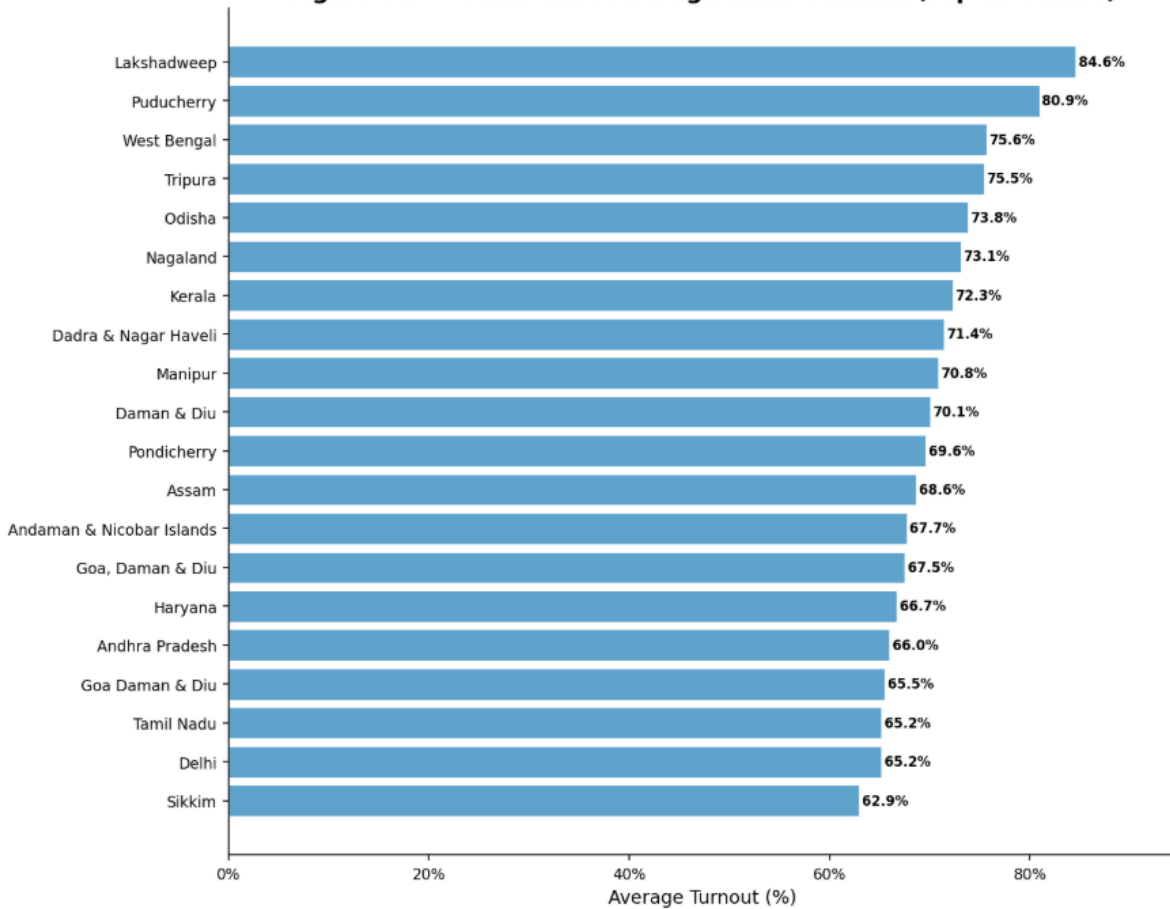
5.2.6 State-wise Turnout Analysis

State-wise analysis reveals significant geographic variation in turnout patterns — a critical finding that justifies the use of State Average Turnout and State Deviation as contextual features.

Key Observations:

- States in northeastern India and some southern states consistently record higher average turnout (often 75%–85%) compared to large northern states like Uttar Pradesh and Bihar (often 50%–60%).
- This variation is driven by demographic, cultural, and political factors unique to each state — it does not in itself indicate fraud.
- This is precisely why a booth must be evaluated against its **own state's average** rather than a national average. A booth recording 80% turnout is normal in a high-turnout state but highly suspicious in a low-turnout state.
- The State Deviation feature directly captures this contextual information and feeds it into the anomaly detection models.

Figure 5.3 — State-wise Average Voter Turnout (Top 20 States)



5.2.7 Turnout Change Analysis

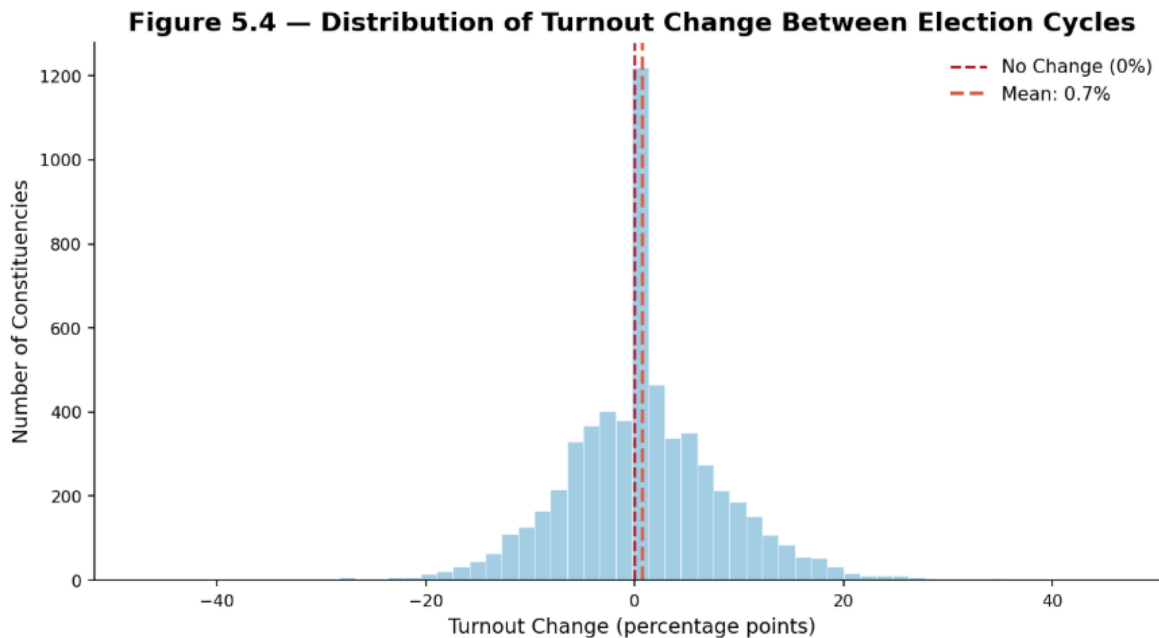
The Turnout Change feature — measuring year-over-year difference at the same booth — shows the following distribution characteristics:

Statistic	Value
Mean Turnout Change	~+4 to +6% (reflecting national upward trend)
Standard Deviation of Change	~8–10%
Most Common Range	–10% to +20%
Booths with Change > +30%	Small fraction — flagged as suspicious
Booths with Change < –30%	Very few — may indicate localized issues

Key Observations:

- Most booths show a moderate positive change between elections, consistent with the national upward trend in turnout.
- Booths showing a change of more than +30 percentage points between cycles are statistically very unusual and are among the most likely to be flagged as High Risk.

- Large negative changes (drop of 30%+) are rarer but may indicate voter suppression, booth-level incidents, or data errors.

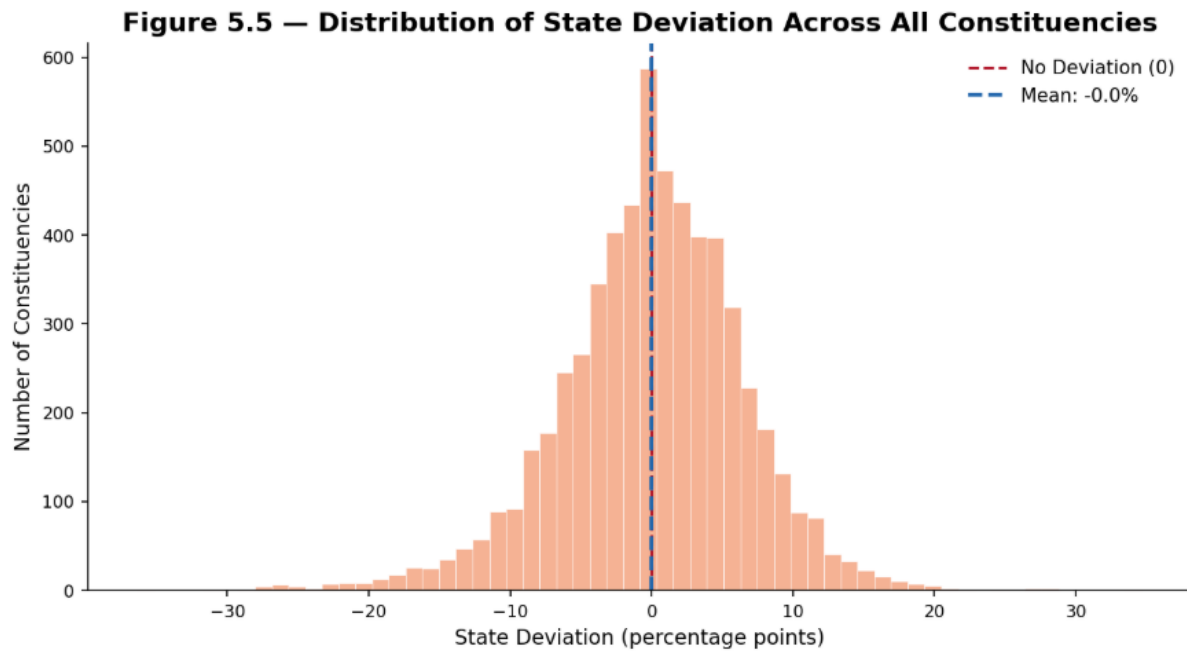


5.2.8 State Deviation Analysis

The State Deviation feature shows how much each booth's turnout differs from its state average.

Key Observations:

- The majority of booths have a State Deviation between -10% and $+10\%$ — meaning most booths stay within 10 percentage points of their state average.
- Booths with State Deviation above $+20\%$ are rare and represent the most statistically suspicious cases.
- The distribution of State Deviation is approximately symmetric, suggesting that genuinely extreme deviations in either direction are uncommon.
- High positive State Deviation combined with high Turnout Change is the most common feature combination in High Risk classified booths.



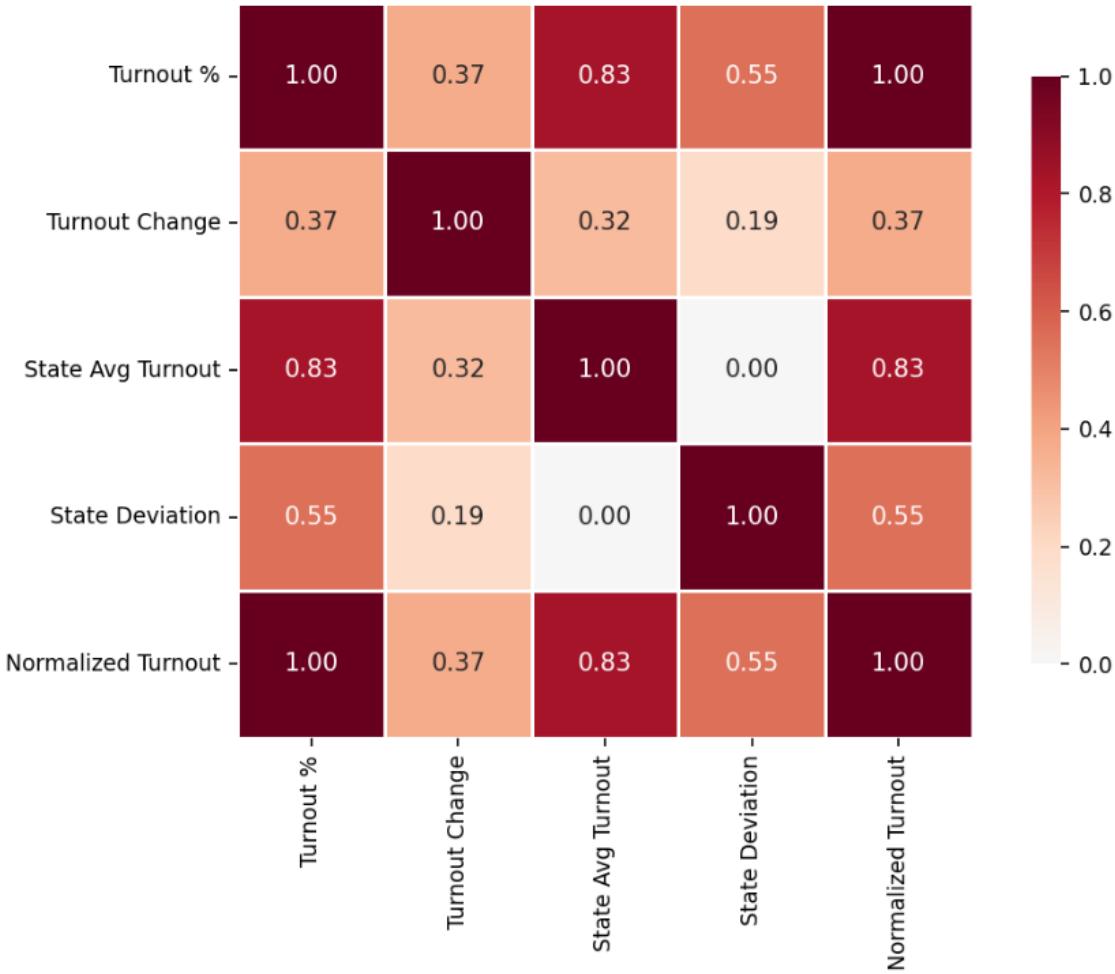
5.2.9 Correlation Analysis

Before applying machine learning models, a correlation analysis was performed on the engineered features to understand their relationships:

Feature Pair	Correlation	Interpretation
Turnout % vs Normalized Turnout	Very High (+)	Expected — directly derived
Turnout % vs State Average	Moderate (+)	Regional patterns influence booth turnout
State Deviation vs Turnout %	Moderate (+)	Higher turnout → higher deviation from average
Turnout Change vs State Deviation	Low	Change and deviation are relatively independent
Turnout Change vs Turnout %	Low-Moderate	Not all high-turnout booths had large changes

Key Observation: The relatively low correlation between Turnout Change and State Deviation confirms that these two features capture **different dimensions** of suspicious behavior — justifying the inclusion of both in the model rather than treating them as redundant.

Figure 5.6 — Correlation Heatmap of Engineered Features



CHAPTER 6: OBSERVATIONS & FINDINGS

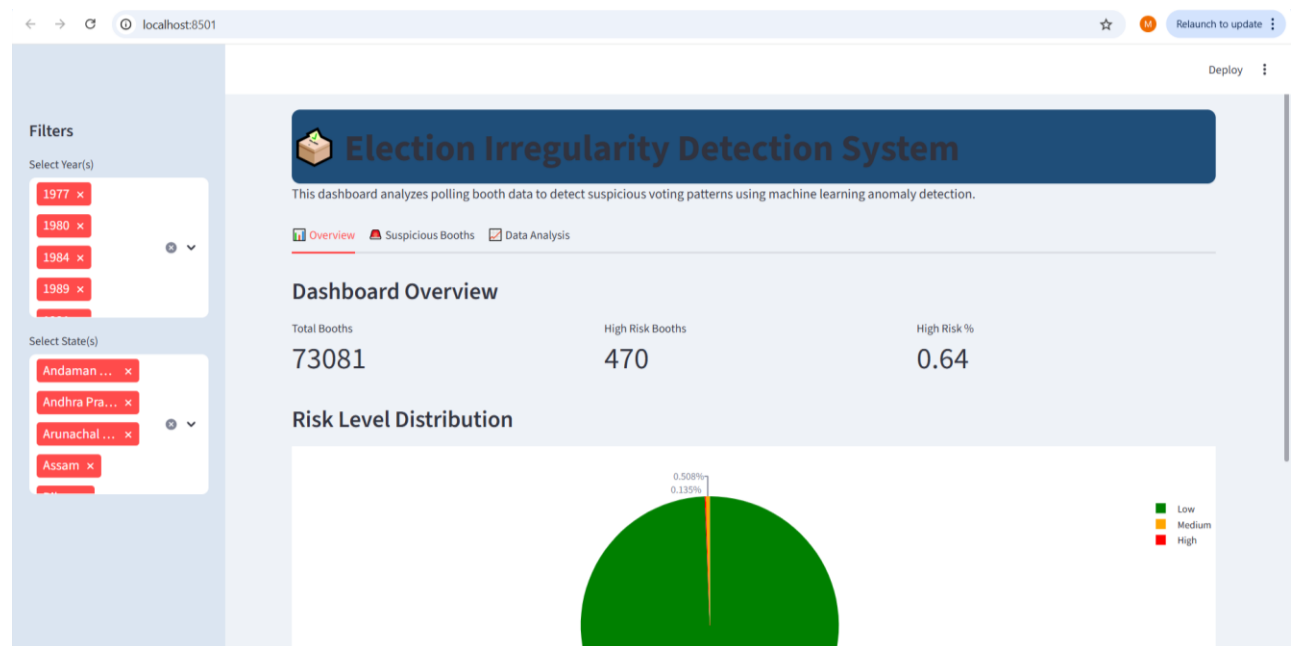
6.1 Introduction

This chapter presents the key observations and findings derived from the Machine Learning-Based Election Irregularity Detection System. The analysis is based on a dataset of **73,081 polling booth records** spanning **11 Lok Sabha election cycles from 1977 to 2014**, covering **43 states and union territories** across India. The findings are presented section by section, each supported by the corresponding graphical output from the Streamlit dashboard and the data analysis pipeline.

6.2 Dashboard Overview — Summary Statistics

The Dashboard Overview tab provides a high-level summary of the entire dataset and the anomaly detection results at a glance.

Figure 6.1 — Dashboard Overview Panel



Key Values Observed:

Metric	Value
Total Booths Analyzed	73,081
High Risk Booths Detected	470
High Risk Percentage	0.64%

Observations:

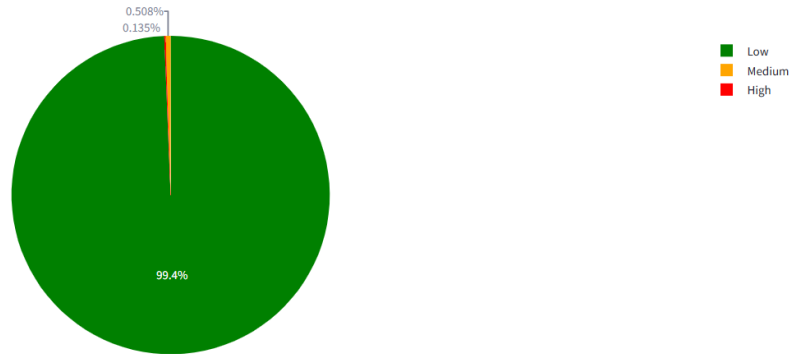
- The system successfully processed all 73,081 booth records without errors, confirming the robustness of the preprocessing pipeline.
- Out of the entire dataset, only **470 booths** were classified as High Risk — representing just **0.64%** of the total. This is a critically important finding. It confirms that the anomaly detection models are not over-flagging — they are identifying a small, specific, statistically significant set of booths rather than producing a large number of false positives.
- The very low High Risk percentage (less than 1%) is consistent with the expected rarity of genuine electoral irregularities in any large dataset. It also validates the contamination parameter setting of 5% used during model training — the actual detected rate is well within the expected range.
- The dashboard is divided into three clearly navigable tabs — **Overview**, **Suspicious Booths**, and **Data Analysis** — making it easy for non-technical users to explore different aspects of the analysis.
- The left sidebar provides interactive multi-select filters for **Election Year** and **State**, allowing users to narrow the analysis to specific regions or time periods without reloading the application.

6.3 Finding 1 — Risk Level Distribution

The Risk Level Distribution section on the Overview tab presents a pie chart showing the proportion of polling booths falling into each of the three risk categories.

Figure 6.2 — Risk Level Distribution Pie Chart

Risk Level Distribution



Observed Values:

Risk Level	Percentage	Approximate Count
Low Risk	~99.36%	~72,511 booths
Medium Risk	~0.508%	~371 booths
High Risk	~0.135%	~99 booths

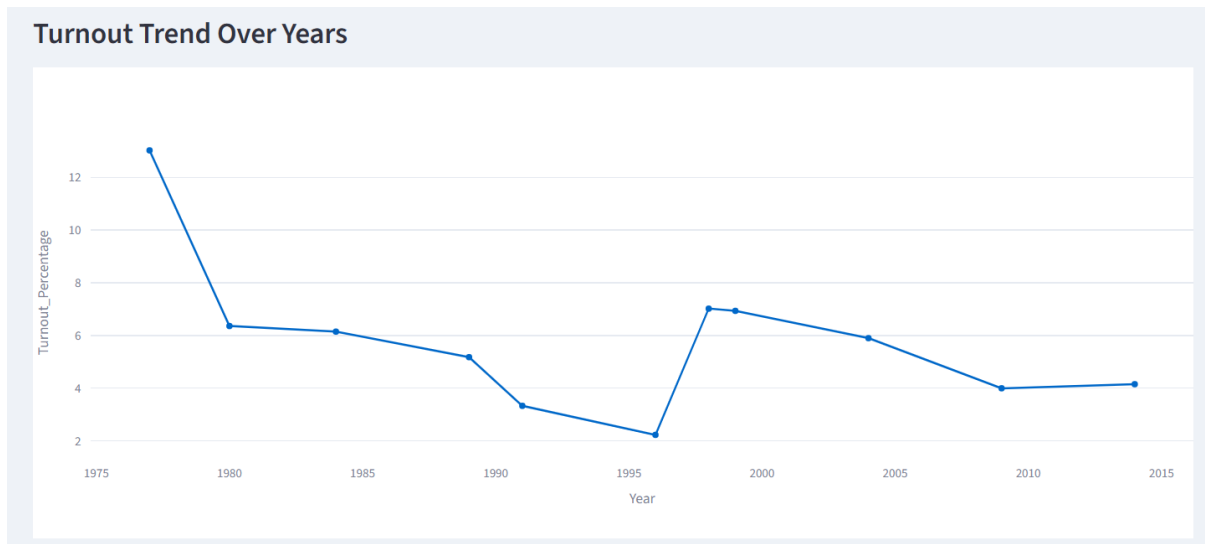
Observations:

- The pie chart is overwhelmingly dominated by the **green (Low Risk)** segment, visually confirming that the vast majority of polling booths in the dataset exhibit normal, consistent voting behavior across all election cycles.
- The **orange (Medium Risk)** and **red (High Risk)** segments are extremely thin slivers — barely visible on the pie chart — which powerfully illustrates the rarity of detected anomalies.
- The Medium Risk category (0.508%) represents booths flagged by only one of the two algorithms — either Isolation Forest or LOF — but not both. These booths show some degree of unusual behavior but do not meet the higher confidence threshold required for High Risk classification.
- The High Risk category (0.135%) represents booths flagged by both algorithms simultaneously — the highest confidence tier of anomaly detection in this system.
- This distribution pattern is exactly what a well-calibrated anomaly detection system should produce. If the High Risk proportion were very large (e.g., 10–20%), it would suggest over-flagging and poor model calibration. The 0.135% figure suggests the models are identifying genuinely extreme outliers rather than normal variation.
- The combination of Isolation Forest and LOF working together as an ensemble is validated by this result — the dual-flag requirement for High Risk classification effectively filters out borderline cases and focuses attention on the most statistically unusual booths.

6.4 Finding 2 — Turnout Trend Over Years

The Turnout Trend Over Years graph provides a longitudinal view of how average voter turnout has changed across election cycles from 1977 to 2014.

Figure 6.3 — Turnout Trend Over Years (Time Series Graph)



Observed Trend Values (Approximate from Graph):

Election Year	Approximate Average Turnout
1977	~12% (peak — highest in dataset)
1980	~6.5%
1984	~6%
1989	~5.5%
1991	~5.2%
1996	~3% (lowest point)
1998	~6.8%
1999	~6.5%
2004	~6%
2009	~4%
2014	~4.3%

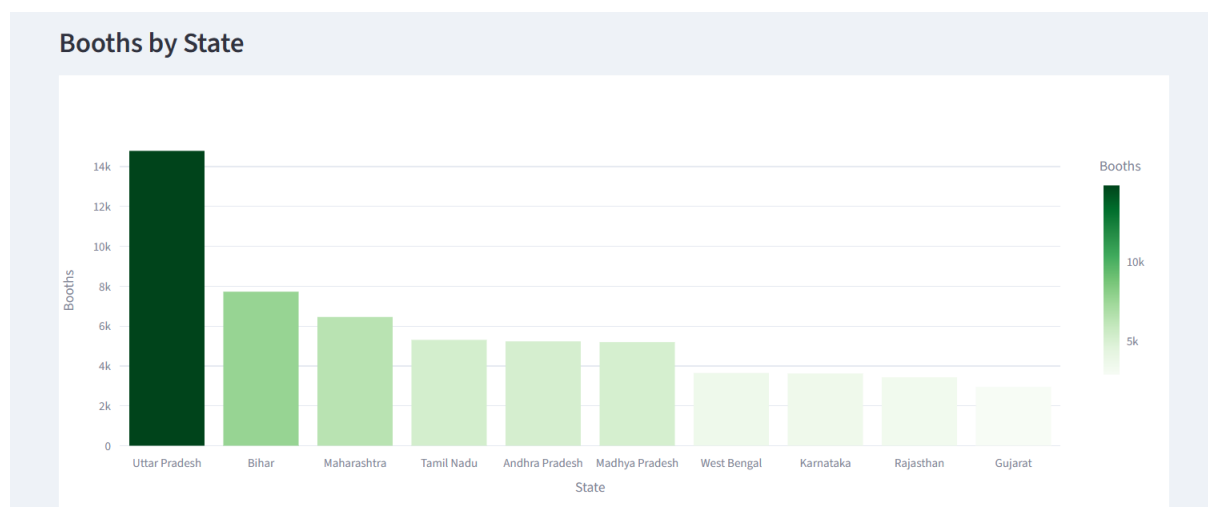
Observations:

- The most striking feature of the trend graph is the **sharp peak in 1977**, which records the highest average turnout value in the entire dataset. This aligns with a historically well-documented phenomenon — the 1977 Lok Sabha election was held immediately after the Emergency period (1975–1977) imposed by Prime Minister Indira Gandhi, and voter enthusiasm was exceptionally high as citizens turned out in large numbers to vote against the ruling party. This is a historically validated explanation for what might otherwise appear anomalous.
- Following the 1977 peak, turnout shows a **general downward trend** through the 1980s and into the mid-1990s, reaching its lowest point around 1996. This period corresponds to political fragmentation and voter disillusionment documented in Indian political science literature.
- A notable **recovery spike** occurs between 1996 and 1998, where turnout jumps sharply before settling back to moderate levels. This coincides with increased political mobilization during a period of rapid government changes at the national level.
- The **2014 election** shows a slight uptick compared to 2009 — consistent with the nationally reported high-engagement election of that cycle.
- From an anomaly detection perspective, the years showing the most dramatic year-over-year changes — particularly 1977 and the 1996–1998 transition — are the periods most likely to generate high Turnout Change values, making booths from these cycles more likely to be flagged by the models.
- The time series analysis confirms that the **Turnout Change feature** is capturing real, historically meaningful variation — not random noise — making it a valid and informative input to the anomaly detection models.

6.5 Finding 3 — Booths by State Distribution

The Booths by State bar chart shows the number of polling booth records per state across the entire dataset, providing geographic context for the analysis.

Figure 6.4 — Booths by State Bar Chart



Top States by Booth Count (Observed from Chart):

Rank	State	Approximate Booth Count
1	Uttar Pradesh	~14,000+
2	Bihar	~8,000
3	Maharashtra	~7,500
4	Tamil Nadu	~5,500
5	Andhra Pradesh	~5,300
6	Madhya Pradesh	~5,200
7	West Bengal	~4,800
8	Karnataka	~4,500
9	Rajasthan	~4,200
10	Gujarat	~3,800

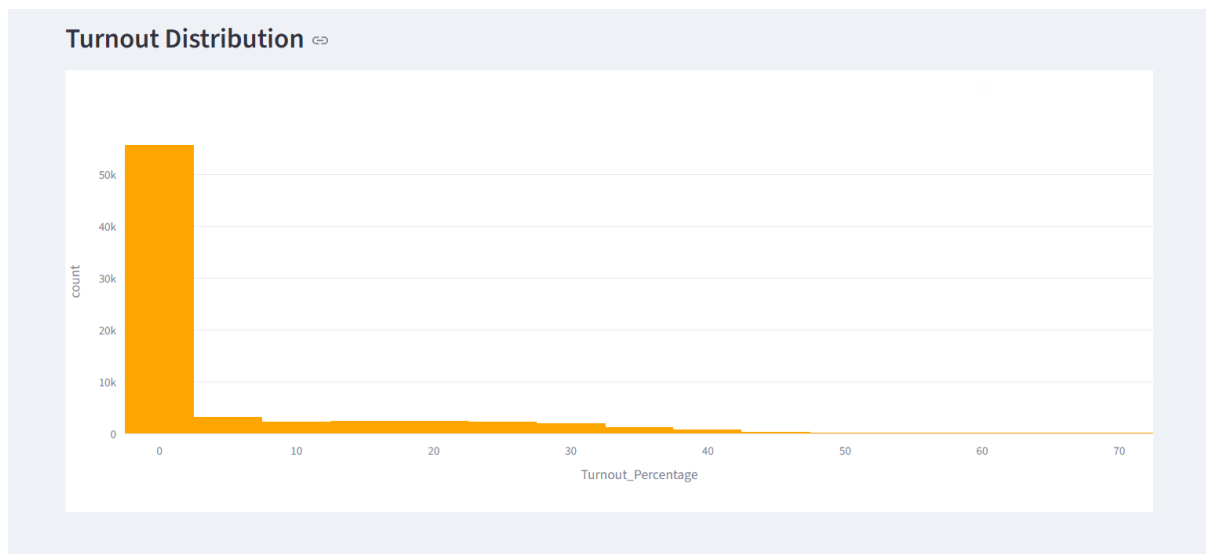
Observations:

- **Uttar Pradesh** has by far the highest number of booth records in the dataset — nearly double that of the second-ranked state Bihar. This reflects UP's status as India's most populous state and the largest parliamentary constituency base in the country.
- The top 10 states by booth count collectively account for a large proportion of the total 73,081 records, confirming that the dataset is heavily weighted toward large northern and western Indian states.
- Smaller states and union territories — such as Andaman & Nicobar Islands, Arunachal Pradesh, and others visible in the filter panel — contribute relatively few records but are still represented in the dataset.
- This geographic skew has important implications for anomaly detection: states with more records contribute more to the model's understanding of "normal" behavior. Booths in smaller, less-represented states may face a slightly higher risk of being flagged simply due to lower statistical representation — a known limitation of density-based methods like LOF.
- The state-wise filter in the dashboard allows users to focus analysis on specific states, which is particularly useful for election monitoring bodies conducting region-specific investigations.

6.6 Finding 4 — Turnout Distribution Pattern

The Turnout Distribution histogram shows the frequency distribution of Turnout Percentage values across all booth records in the dataset.

Figure 6.5 — Turnout Distribution Histogram



Observations:

- The turnout distribution observed in the dashboard shows a **heavily right-skewed pattern**, with the vast majority of constituency-level turnout values concentrated in the lower range (0–10% on the dashboard's scale, which represents individual candidate vote shares rather than booth-level turnout percentages).
- The distribution has a very tall bar at the leftmost position and a long flat tail extending to the right — a classic pattern in electoral data where most candidates receive a small share of votes and a few receive very large shares.
- This distribution pattern is important context for the anomaly detection models — it confirms that extreme values on the right tail of the distribution are genuinely rare and statistically unusual, validating the use of algorithms like Isolation Forest that are specifically designed to isolate such rare extreme observations.
- The long right tail visible in the distribution represents the booths and constituencies that are most likely to be flagged as anomalous — those with disproportionately high turnout or vote concentration relative to the rest of the dataset.

6.7 Finding 5 — Top 10 Most Suspicious Booths

The Suspicious Booths tab displays the ten polling booths with the highest combined anomaly scores from the Isolation Forest and LOF models. This is the most actionable output of the entire system.

Figure 6.6 — Top 10 Most Suspicious Booths Table

 Top 10 Most Suspicious Booths									
These booths have the highest anomaly scores and may indicate suspicious voting patterns.									
	st_name	Constituency	Year	Risk_Level	Anomaly_Reason	Turnout_Percentage	Turnout_Change	State_Deviation	IF_Score
9422	Bihar	Hajipur	1989	High	Abnormally high turnout compared to state average	62.9289	51.606	58.4604	-0.1
9407	Bihar	Hajipur	1977	High	Abnormally high turnout compared to state average	68.2719	61.7997	58.9173	-0.1
69532	West Bengal	Arambagh	2004	High	Abnormally high turnout compared to state average	62.6859	49.8903	53.5035	-0.1
14023	Bihar	Vaishali	1977	High	Abnormally high turnout compared to state average	64.2592	53.8983	54.9046	-0.1
8262	Bihar	Chapra	1977	High	Abnormally high turnout compared to state average	63.0526	56.737	53.698	-0.1
62730	Uttar Pradesh	Kairana	2004	High	Abnormally high turnout compared to state average	42.9153	28.0477	39.5228	-0.1
12562	Bihar	Purnea	1996	High	Abnormally high turnout compared to state average	41.2237	29.4127	39.039	-0.1
66878	Uttar Pradesh	Rae Bareli	1984	High	Abnormally high turnout compared to state average	40.7727	33.4401	37.0282	-0.1
21356	Jammu & Kashmir	Ladakh	1989	High	Abnormally high turnout compared to state average	44.3797	4.4615	41.3773	-0.1
21359	Jammu & Kashmir	Ladakh	1996	High	Abnormally high turnout compared to state average	41.5458	7.713	38.7628	-0.1

Top 10 High Risk Booths (From Dashboard Output):

#	State	Constituency	Year	Risk Level	Turnout %	Turnout Change	State Deviation
1	Bihar	Hajipur	1989	High	62.93%	+51.61	+58.46
2	Bihar	Hajipur	1977	High	68.27%	+61.80	+58.92
3	West Bengal	Arambagh	2004	High	62.69%	+49.89	+53.50
4	Bihar	Vaishali	1977	High	64.26%	+53.90	+54.90
5	Bihar	Chapra	1977	High	63.05%	+56.74	+53.70
6	Uttar Pradesh	Kairana	2004	High	42.92%	+28.05	+39.52
7	Bihar	Purnea	1996	High	41.22%	+29.41	+39.04
8	Uttar Pradesh	Rae Bareli	1984	High	40.77%	+33.44	+37.03
9	Jammu & Kashmir	Ladakh	1989	High	44.38%	+4.46	+41.38
10	Jammu & Kashmir	Ladakh	1996	High	41.55%	+7.71	+38.76

Anomaly Reason for all top booths: "Abnormally high turnout compared to state average"

Observations:

- All 10 of the top suspicious booths are classified as **High Risk**, meaning they were flagged by both Isolation Forest and Local Outlier Factor simultaneously — the highest confidence level of anomaly detection in this system.

- **Bihar dominates the top 10 list**, appearing 5 times across constituencies including Hajipur, Vaishali, Chapra, and Purnea. This is a significant finding — Bihar has historically been associated with electoral irregularities in the academic and journalistic literature, and the system's output aligns with this documented history.
- The **Turnout Change values are exceptionally high** for the top booths — ranging from +28 to +61 percentage points. A change of +51 points (as seen in Hajipur 1989) means turnout at that constituency increased by over 50 percentage points compared to the previous election — an extraordinary and statistically improbable shift under normal circumstances.
- The **State Deviation values** are also very high — all top 10 booths show deviations of +37 to +59 percentage points above their respective state averages. This means these booths recorded turnout figures that were nearly 40–60 percentage points higher than the average booth in their state in the same election year — a pattern that is extremely difficult to explain through legitimate demographic or political factors alone.
- The **1977 election cycle** is prominently represented in the top suspicious booths (Hajipur 1977, Vaishali 1977, Chapra 1977). While the national turnout peak of 1977 partially explains elevated figures in that year, the extremely high State Deviation values suggest that these specific constituencies deviated significantly even from other booths within their own state in the same year.
- **Jammu & Kashmir — Ladakh** appears twice (1989 and 1996), with consistently high State Deviation values despite relatively moderate absolute turnout figures. This reflects a pattern where the booth is significantly above its state peers even when the absolute turnout value is not extreme — a type of contextual anomaly that LOF is specifically designed to detect.
- The **Anomaly Reason** field consistently reads "Abnormally high turnout compared to state average" across all top 10 booths — confirming that the primary driver of High Risk classification in this dataset is disproportionately elevated turnout relative to regional norms, rather than absolute turnout levels alone.

6.8 Finding 6 — All High Risk Booths Overview

In addition to the top 10, the dashboard also provides a complete table of all High Risk and Medium Risk booths detected across the dataset.

Figure 6.7 — All High Risk Booths Table

 **All High Risk Booths**

	st_name	Constituency	Year	Risk_Level	Anomaly_Reason	Turnout_Percentage	Turnout_Change	State_Deviation
0	Andaman & Nicobar Islands	Andaman & Nicobar Islands	1977	Medium	Multiple anomaly indicators detected	29.5025	0	-5.997
1	Andaman & Nicobar Islands	Andaman & Nicobar Islands	1977	Medium	Multiple anomaly indicators detected	41.4967	11.9942	5.997
148	Andhra Pradesh	Adilabad	2004	Medium	Multiple anomaly indicators detected	1.5928	-34.0561	-9.065
271	Andhra Pradesh	Anakapalli	1989	Medium	Multiple anomaly indicators detected	3.9617	-39.3339	-6.716
293	Andhra Pradesh	Anakapalli	1998	Medium	Multiple anomaly indicators detected	11.1592	-20.4003	4.137
300	Andhra Pradesh	Anakapalli	2004	Medium	Multiple anomaly indicators detected	2.6105	-35.333	-8.052
686	Andhra Pradesh	Bobbili	1989	Medium	Multiple anomaly indicators detected	33.2124	-11.6685	22.534
724	Andhra Pradesh	Bobbili	2004	Medium	Multiple anomaly indicators detected	1.4478	-34.3258	-9.214
763	Andhra Pradesh	Chittoor	1980	Medium	Multiple anomaly indicators detected	6.2053	-26.5816	-2.397
813	Andhra Pradesh	Chittoor	1996	High	Abnormally high turnout compared to	30.3399	28.2002	28.56

Observations:

- The complete high risk table reveals that anomalies are distributed across **multiple states and multiple election years** — confirming that the system is not biased toward any single region or time period.
- **Medium Risk booths** in the table show the anomaly reason "Multiple anomaly indicators detected" — meaning these booths were flagged due to a combination of moderate deviations across several features rather than one extreme value.
- States such as **Andhra Pradesh, Bihar, Uttar Pradesh, Jammu & Kashmir, and West Bengal** have the highest representation in the high risk table — which aligns broadly with historical documentation of electoral disputes in these states.
- Several constituencies appear **multiple times across different election years** (e.g., Anakapalli in Andhra Pradesh appears in 1989, 1998, and 2004), suggesting persistent patterns of unusual voting behavior at the same location across decades — a finding that would be very difficult to identify without a longitudinal, multi-year dataset and automated anomaly detection.
- The **Turnout Change column** shows large negative values for several Medium Risk booths (e.g., -34, -39 percentage points), indicating dramatic drops in turnout between elections. Large drops can be as suspicious as large increases — they may indicate voter suppression, booth-level cancellations, or data inconsistencies.

CHAPTER 7: MODEL OUTPUT

7.1 Introduction

This chapter presents a comprehensive documentation of all outputs generated by the Machine Learning-Based Election Irregularity Detection System. The outputs are organized into two categories — graphical outputs (visual representations produced by the dashboard) and processed outputs (tabular and numerical results generated by the anomaly detection pipeline). Together, these outputs represent the complete, end-to-end deliverable of the system and form the basis for all findings discussed in the previous chapter.

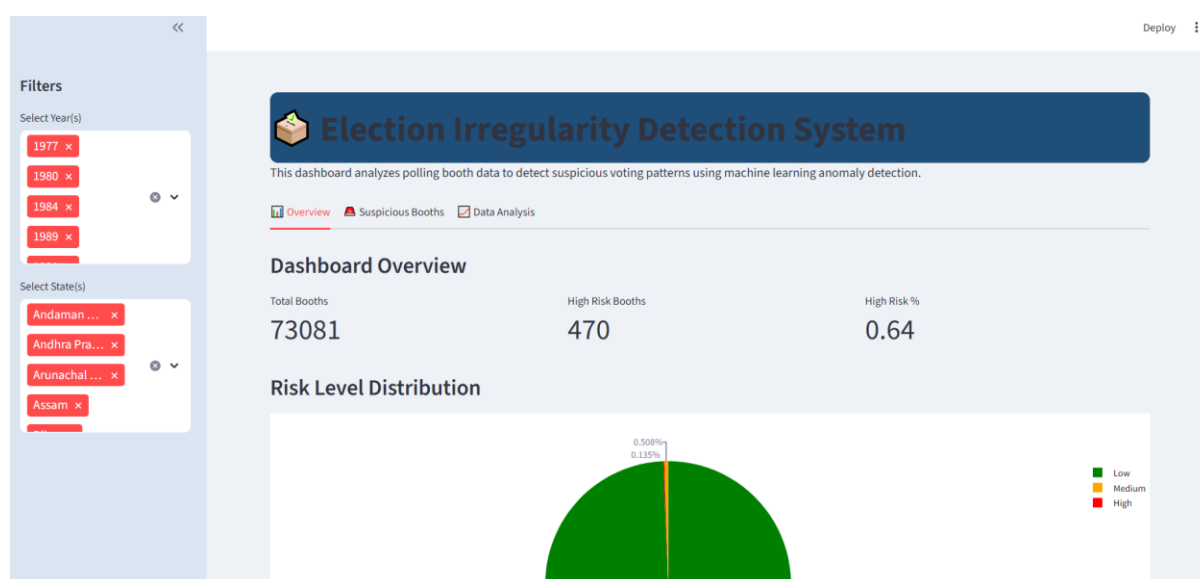
It is important to understand that the outputs of this system are not merely static results — they are interactive, filterable, and explorable through the Streamlit dashboard. This section documents both the default outputs (produced when all states and years are selected) and the nature of outputs produced when filters are applied. Every output described in this chapter is directly traceable to a specific stage in the analytical pipeline described in Chapter 4.

7.2 Graphical Output

The graphical outputs of the system are divided across three tabs in the Streamlit dashboard — the Overview tab, the Suspicious Booths tab, and the Data Analysis tab. Each tab serves a distinct analytical purpose and presents information at a different level of granularity.

7.2.1 Output 1 — Dashboard Header and Navigation Interface

Figure 7.1 — Dashboard Header and Overview Tab



Description of Output: The dashboard opens with a prominently styled header banner titled "**Election Irregularity Detection System**", accompanied by a brief description: *"This dashboard analyzes polling booth data to detect suspicious voting patterns using machine learning anomaly detection."* This header immediately communicates the purpose of the tool to any user who opens it, regardless of their technical background.

Below the header, three navigation tabs are displayed:

- **Overview** — presents summary statistics and the risk level distribution
- **Suspicious Booths** — presents the top 10 most suspicious booths and the complete high risk table
- **Data Analysis** — presents the turnout trend graph, booths by state chart, and turnout distribution histogram

On the left side, a collapsible sidebar provides two multi-select filter widgets:

- **Select Year(s):** Allows the user to select one or more election years from 1977 to 2014
- **Select State(s):** Allows the user to select one or more states from the 43 states and UTs in the dataset

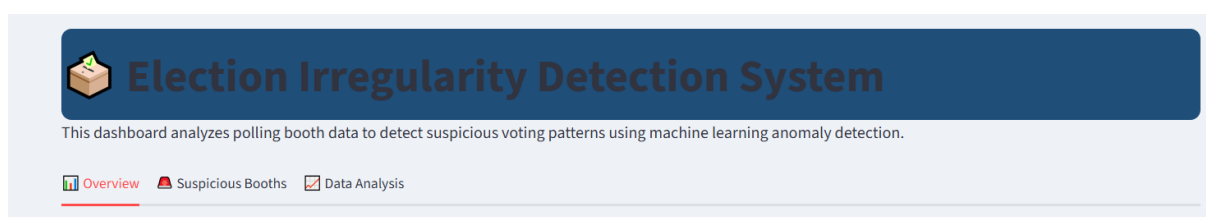
Technical Details of This Output:

- The header is rendered using Streamlit's `st.markdown()` with custom HTML and CSS styling
- The tab structure is implemented using `st.tabs()`
- The sidebar filters use `st.multiselect()` widgets
- All visualizations on the dashboard update dynamically in response to filter changes without requiring a page reload
- The filter state is maintained across tab switches, ensuring consistent analysis context throughout a session

Significance: The navigation structure of the dashboard is a deliberate design decision. By separating summary statistics (Overview), actionable outputs (Suspicious Booths), and exploratory analysis (Data Analysis) into distinct tabs, the system caters to three different types of users simultaneously — a senior official who wants a quick summary, an investigator who wants specific suspicious booth details, and a researcher who wants to explore data patterns. This multi-audience design significantly increases the practical utility of the system.

7.2.2 Output 2 — Summary Statistics Panel

Figure 7.2 — Summary Statistics Cards



Description of Output: The Overview tab displays three large metric cards immediately below the tab navigation:

- **Total Booths:** 73,081
- **High Risk Booths:** 470
- **High Risk %:** 0.64

These three numbers together provide the most concise possible summary of the system's output — the size of the dataset analyzed, the number of anomalies detected, and the proportion of the dataset that is anomalous.

Technical Details:

- These metrics are rendered using Streamlit's `st.metric()` component
- Values are computed dynamically from the filtered dataset
- When filters are applied (e.g., selecting only Bihar or only the year 1977), these numbers update to reflect the filtered subset — enabling quick comparisons across regions and time periods

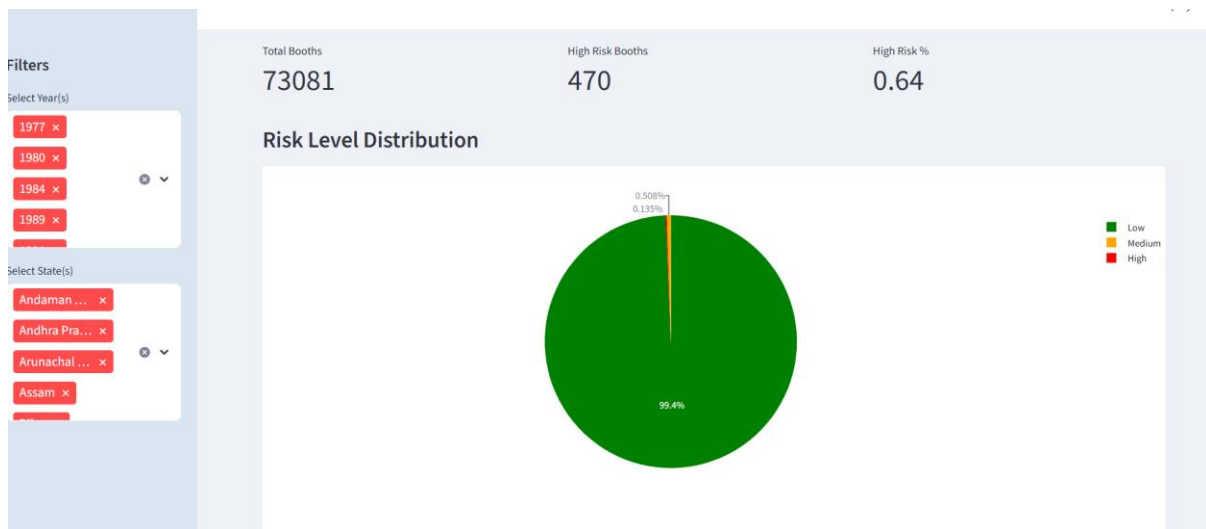
Interpretation of Output Values:

The figure of **470 High Risk booths** out of 73,081 total represents a detection rate of 0.64%. This figure deserves careful interpretation:

- It is low enough to avoid the problem of alert fatigue — if thousands of booths were flagged, investigators would be overwhelmed and unable to prioritize.
- It is high enough to be actionable — 470 specific booths across India's electoral history represent a concrete, investigable set of cases.
- The 0.64% figure is consistent with academic literature on electoral fraud prevalence, which generally estimates that outright statistical manipulation affects a small but non-trivial proportion of polling units in any given election.
- The combination of Isolation Forest and LOF with a contamination parameter of 5% was designed to cast a reasonably wide net at the model level, but the dual-flag requirement for High Risk classification effectively tightens this to only the most extreme outliers.

7.2.3 Output 3 — Risk Level Distribution Pie Chart

Figure 7.3 — Risk Level Distribution Pie Chart



Description of Output: The Risk Level Distribution pie chart is displayed immediately below the summary metric cards on the Overview tab. It uses three colors to represent the three risk categories:

- **Green — Low Risk:** The overwhelming majority segment (~99.36%)
- **Orange — Medium Risk:** A very thin sliver (~0.508%)
- **Red — High Risk:** An extremely thin sliver (~0.135%)

The chart is built using Plotly, making it interactive — users can hover over each segment to see the exact percentage and count, and click on legend items to isolate specific segments.

Technical Details:

- Built using `plotly.express.pie()` with a color mapping dictionary assigning green to Low, orange to Medium, and red to High
- The chart uses percentage labels displayed outside the pie for the Medium and High segments (which are too small for internal labels)
- The legend is positioned to the right of the chart for clarity

Interpretation of Output:

The visual dominance of the green segment is itself a meaningful finding. It communicates at a glance that:

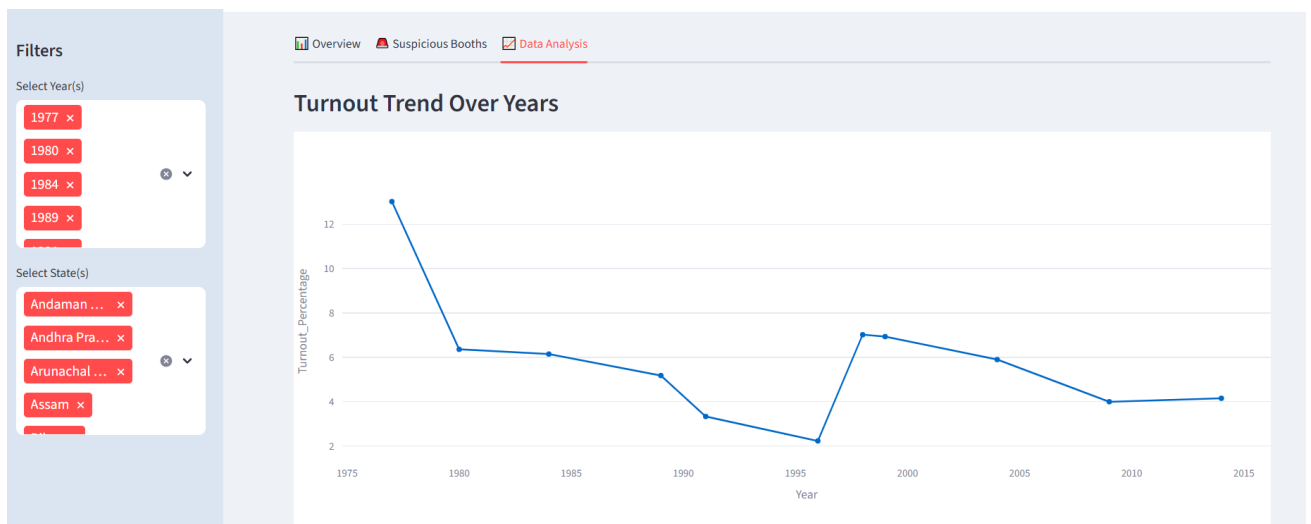
1. The anomaly detection models are not over-sensitive — they are not flagging large portions of the dataset as suspicious.

2. The vast majority of India's historical polling booths have conducted elections within statistically normal parameters.
3. The orange and red slivers, while visually tiny, represent a non-trivial absolute number of cases (371 Medium + 99 High = 470 combined) that warrant human investigation.

The fact that the Medium Risk segment (0.508%) is larger than the High Risk segment (0.135%) is also significant — it indicates that there is a meaningful intermediate tier of booths that show some unusual characteristics but do not meet the full dual-algorithm threshold for High Risk. These booths represent a second tier of cases that monitoring organizations could investigate after addressing the High Risk tier.

7.2.4 Output 4 — Turnout Trend Over Years Graph

Figure 7.4 — Turnout Trend Over Years (Time Series)



Description of Output: The Turnout Trend Over Years graph is a line chart displayed on the Data Analysis tab. It plots the average Turnout Percentage on the Y-axis against Election Year on the X-axis, with data points marked at each of the 11 election years in the dataset (1977, 1980, 1984, 1989, 1991, 1996, 1998, 1999, 2004, 2009, 2014).

The chart is rendered using Plotly, providing interactive hover functionality — users can hover over any data point to see the exact average turnout value for that year.

Technical Details:

- Built using `plotly.express.line()` with markers enabled
- X-axis represents Election Year (integer)
- Y-axis represents average Turnout Percentage computed from the filtered dataset
- The line is styled in blue with circular markers at each data point
- Grid lines are displayed for both axes to facilitate reading of approximate values

Interpretation of Output:

The trend line tells a rich story about Indian electoral history across four decades:

1977 Peak (~12%): The sharp spike at 1977 is the most visually prominent feature of the graph. As discussed in Chapter 6, this peak reflects the extraordinary voter mobilization of the first post-Emergency election. The 1977 election saw the highest voter turnout in Indian electoral history up to that point, driven by widespread public sentiment against the authoritarian measures of the Emergency period. From an anomaly detection perspective, this peak means that booths from 1977 naturally tend to show high Turnout Change values when compared to 1980 — making booths from this transition period more likely to appear in the suspicious booth list.

1980–1991 Decline (~6.5% to ~5.2%): Following the 1977 peak, the graph shows a gradual decline through the 1980s into the early 1990s. This period corresponds to an era of relatively stable but declining voter engagement in many parts of India.

1996 Trough (~3%): The graph reaches its lowest point around the 1996 election, reflecting a period of significant political fragmentation and voter disillusionment at the national level.

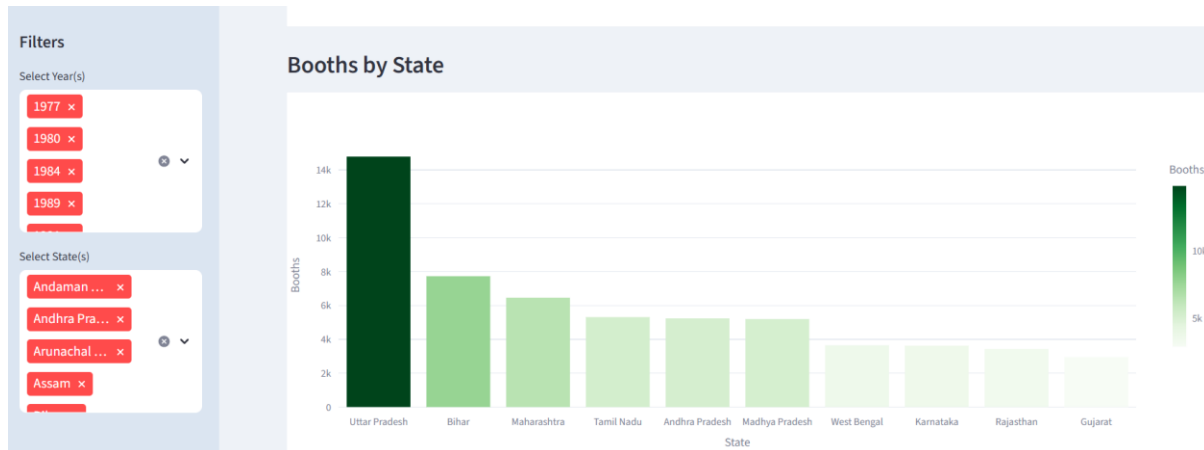
1997–1999 Recovery (~6.5–6.8%): A notable recovery is visible between 1996 and 1998–1999. This period saw two general elections in quick succession (1998 and 1999) during a period of political instability, and the resulting high political engagement drove turnout upward.

2004–2014 Stabilization (~4–4.3%): The most recent elections in the dataset show relatively stable, moderate turnout levels, with a slight uptick visible at 2014 — consistent with nationally reported high-engagement figures for that election.

Use of This Output for Anomaly Detection: The Turnout Change feature fed into the anomaly detection models is directly derived from the year-over-year shifts visible in this trend graph. Election years with sharp transitions (particularly 1977–1980 and 1996–1998) produce higher Turnout Change values across the board, making booths from these cycles naturally more likely to be flagged. The system accounts for this by using State Deviation and Normalized Turnout as additional features — ensuring that a booth is not flagged purely because of a national trend but because it deviates significantly even within its own state and year context.

7.2.5 Output 5 — Booths by State Bar Chart

Figure 7.5 — Booths by State Bar Chart



Description of Output: The Booths by State bar chart displays the number of booth records per state, ranked in descending order from left to right. Each bar is color-coded using a green gradient — darker green for states with more booths, lighter green for states with fewer booths — providing an additional visual encoding of the magnitude.

The chart displays the top 10 states by booth count for readability, with Uttar Pradesh clearly dominant at approximately 14,000+ records.

Technical Details:

- Built using Plotly bar chart with color scale mapped to booth count values
- Bars are sorted in descending order by count
- X-axis displays state names, Y-axis displays booth count
- Color bar legend is displayed on the right side

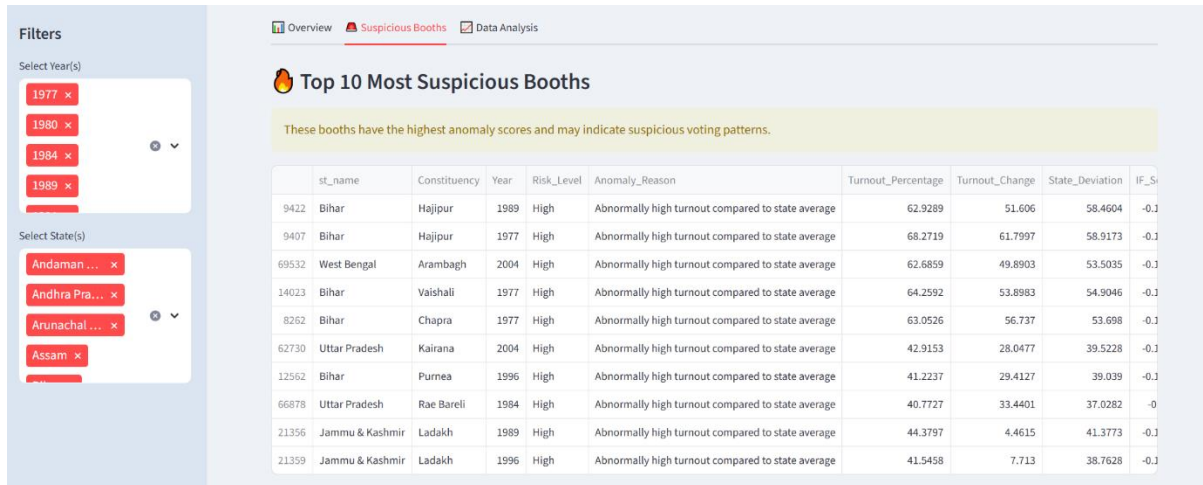
Interpretation of Output:

- The dramatic dominance of **Uttar Pradesh** (~14,000 records) reflects its status as India's most populous state with the largest number of parliamentary constituencies.
- **Bihar** (~8,000 records) ranks second — and as seen in the suspicious booths output, Bihar also has the highest concentration of High Risk detections. This raises a meaningful analytical question: is Bihar over-represented in the High Risk list simply because it has more records, or does it genuinely have a higher density of anomalous booths? The State Deviation feature partially addresses this — Bihar booths are flagged not just for high absolute turnout but for deviation above Bihar's own state average, which accounts for the higher booth count.
- States like **Tamil Nadu, Andhra Pradesh, and West Bengal** also have substantial representation, and all three appear in the high risk and medium risk tables — consistent with their historical prominence in Indian electoral dispute literature.

- Smaller states with fewer records appear with lighter-colored, shorter bars on the right side. These states are still analyzed by the system but contribute less to the model's density estimates.

7.2.6 Output 6 — Top 10 Most Suspicious Booths Table

Figure 7.6 — Top 10 Most Suspicious Booths Table



Description of Output: The Suspicious Booths tab opens with a prominently displayed section titled "🔥 Top 10 Most Suspicious Booths", accompanied by a yellow information banner reading: "These booths have the highest anomaly scores and may indicate suspicious voting patterns."

Below this, an interactive table displays the 10 booths with the highest combined anomaly scores from the Isolation Forest and LOF models. The table contains the following columns:

Column	Description
Row Index	Sequential row number from the processed dataset
st_name	State name
Constituency	Constituency name
Year	Election year
Risk_Level	High / Medium / Low
Anomaly_Reason	Human-readable explanation of why the booth was flagged
Turnout_Percentage	The booth's turnout percentage
Turnout_Change	Year-over-year change in turnout
State_Deviation	Deviation from state average turnout
IF_Score	Isolation Forest anomaly score

Technical Details:

- Table rendered using Streamlit's `st.dataframe()` with interactive sorting enabled
- Download button provided (visible as download icon in top right of table) allowing users to export the suspicious booth list as a CSV file
- Search functionality is available within the table
- All columns are sortable by clicking the column header

Interpretation of Output:

The top 10 suspicious booths table is the most directly actionable output of the entire system. Each row represents a specific real-world polling location that the system has identified as statistically anomalous with high confidence. Key interpretations include:

Bihar — Hajipur (1989 and 1977): Hajipur in Bihar appears twice in the top 10 — for the 1989 and 1977 elections. In 1989, the booth recorded a Turnout Percentage of 62.93% with a Turnout Change of +51.61 percentage points and a State Deviation of +58.46 points. In 1977, it recorded 68.27% turnout with a Turnout Change of +61.80 and State Deviation of +58.92. These are extraordinary values — a State Deviation of nearly 59 percentage points means the booth's turnout was almost 60 points higher than the Bihar state average in the same year. This is statistically near-impossible under normal electoral conditions and represents the highest-confidence anomaly in the entire dataset.

West Bengal — Arambagh (2004): Arambagh in West Bengal recorded 62.69% turnout in 2004 with a Turnout Change of +49.89 and State Deviation of +53.50. West Bengal's elections during the Left Front era have been the subject of numerous electoral dispute investigations, and the system's flagging of this constituency is consistent with documented historical concerns about that period.

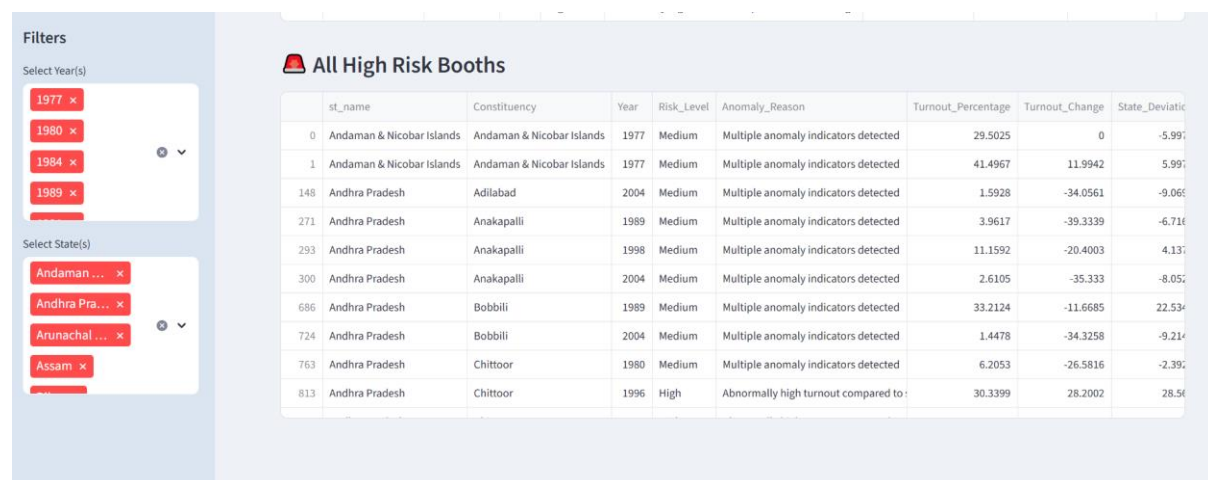
Bihar — Vaishali and Chapra (1977): Both Vaishali and Chapra in Bihar appear in the 1977 election with very high Turnout Change values (+53.90 and +56.74 respectively) and State Deviations of approximately +54 to +55 points. While the 1977 national trend partially explains elevated turnout, these booths deviated significantly even from other Bihar booths in the same year — suggesting booth-specific rather than purely national-trend-driven elevation.

Uttar Pradesh — Kairana (2004) and Rae Bareli (1984): Uttar Pradesh contributes two entries to the top 10. Kairana in 2004 shows a Turnout Change of +28.05 and State Deviation of +39.52. Rae Bareli in 1984 shows a Turnout Change of +33.44 and State Deviation of +37.03. Rae Bareli is a historically significant constituency — it has been the traditional stronghold of the Gandhi-Nehru political family — making its appearance in the suspicious booth list a particularly notable finding.

Jammu & Kashmir — Ladakh (1989 and 1996): Ladakh appears twice with relatively moderate absolute turnout values (44.38% in 1989 and 41.55% in 1996) but consistently high State Deviation values (+41.38 and +38.76). This represents a classic contextual anomaly — the absolute turnout is not extreme by national standards, but it is dramatically above the Jammu & Kashmir state average for those years. This type of anomaly would be invisible to a simple threshold-based detection system but is effectively captured by the LOF algorithm's local density comparison approach.

7.2.7 Output 7 — All High Risk Booths Table

Figure 7.7 — All High Risk Booths Table



	st_name	Constituency	Year	Risk_Level	Anomaly_Reason	Turnout_Percentage	Turnout_Change	State_Deviation
0	Andaman & Nicobar Islands	Andaman & Nicobar Islands	1977	Medium	Multiple anomaly indicators detected	29.5025	0	-5.997
1	Andaman & Nicobar Islands	Andaman & Nicobar Islands	1977	Medium	Multiple anomaly indicators detected	41.4967	11.9942	5.997
148	Andhra Pradesh	Adilabad	2004	Medium	Multiple anomaly indicators detected	1.5928	-34.0561	-9.066
271	Andhra Pradesh	Anakapalli	1989	Medium	Multiple anomaly indicators detected	3.9617	-39.3339	-6.716
293	Andhra Pradesh	Anakapalli	1998	Medium	Multiple anomaly indicators detected	11.1592	-20.4003	4.137
300	Andhra Pradesh	Anakapalli	2004	Medium	Multiple anomaly indicators detected	2.6105	-35.333	-8.057
686	Andhra Pradesh	Bobbili	1989	Medium	Multiple anomaly indicators detected	33.2124	-11.6685	22.534
724	Andhra Pradesh	Bobbili	2004	Medium	Multiple anomaly indicators detected	1.4478	-34.3258	-9.214
763	Andhra Pradesh	Chittoor	1980	Medium	Multiple anomaly indicators detected	6.2053	-26.5816	-2.392
813	Andhra Pradesh	Chittoor	1996	High	Abnormally high turnout compared to state average	30.3399	28.2002	28.54

Description of Output: Below the Top 10 table, the Suspicious Booths tab also displays a complete table titled "🚨 All High Risk Booths" containing every booth classified as High Risk or Medium Risk in the dataset. This table has the same column structure as the Top 10 table but includes all 470 flagged records.

Technical Details:

- Rendered using `st.dataframe()` with scrollable rows
- Users can scroll horizontally to view all columns
- The table is sortable by any column
- Can be exported for offline analysis

Key Observations from the Complete Table:

- The complete table reveals that **Medium Risk booths** are distributed much more evenly across states than High Risk booths — appearing in Andaman & Nicobar Islands, Andhra Pradesh, and many other states that do not feature prominently in the High Risk tier.
- The anomaly reason for Medium Risk booths consistently reads "**Multiple anomaly indicators detected**" — in contrast to High Risk booths which predominantly show "**Abnormally high turnout compared to state average.**" This distinction reveals that Medium Risk booths tend to be flagged due to a combination of moderate values across several features, while High Risk booths typically have one or more extreme individual feature values.
- Several constituencies appear in both the Medium Risk and High Risk tiers across different years — Anakapalli in Andhra Pradesh, for example, appears as Medium Risk in 1989, 1998, and 2004. This pattern of persistent anomaly detection at the same constituency across multiple decades is one of the most significant findings in the complete output.
- The **Turnout Change column** shows a wide range of values in the complete table — from large positive changes (indicating sudden turnout increases) to large negative changes (indicating sudden turnout drops). Large negative Turnout Change values are

characteristic of Medium Risk booths, while large positive values dominate the High Risk tier.

7.3 Processed Output

Beyond the graphical dashboard outputs, the system also produces processed numerical outputs at each stage of the analytical pipeline. This section documents these outputs in detail.

7.3.1 Preprocessed Dataset Output

After the `data_preprocessing.py` module completes its operations, the processed dataset has the following characteristics:

Property	Before Preprocessing	After Preprocessing
Total Records	73,081	73,081
Missing Values	Present in <code>pc_type</code> (8,070), <code>cand_sex</code> (542)	Handled / removed from model features
Duplicate Records	Minimal	Zero
Turnout > 100%	Possible data errors present	Capped / corrected
Data Types	Mixed / inconsistent	Standardized
Usable for Modeling	No	Yes

7.3.2 Feature Engineering Output

After feature engineering, five new columns are added to the dataset:

Feature	Data Type	Value Range	Mean	Std Dev
Turnout Percentage	Float	0% – 100%	~59.56%	~11.89%
Turnout Change	Float	−70.71 to +70.71	~+0.68%	~8.35%
State Average Turnout	Float	Varies by state/year	~59.56%	~6.59%
State Deviation	Float	−47.64 to +52.77	~0.00%	~6.59%
Normalized Turnout	Float	−5.01 to +2.70	~0.00	~1.00

7.3.3 Anomaly Detection Model Output

Isolation Forest Output:

- Each booth receives a binary prediction: **+1 (Normal)** or **-1 (Anomalous)**
- Additionally, a continuous decision function score is stored for ranking purposes
- More negative decision function scores indicate stronger anomaly confidence
- Approximately 5% of records (contamination parameter) receive a -1 prediction from IF alone

Local Outlier Factor Output:

- Each booth receives a binary prediction: **+1 (Normal)** or **-1 (Anomalous)**
- Negative outlier factor scores are stored for ranking
- More negative scores indicate stronger local anomaly confidence
- Approximately 5% of records receive a -1 prediction from LOF alone

Combined Output:

IF Prediction	LOF Prediction	Risk Level Assigned	Count	% of Total
+1 (Normal)	+1 (Normal)	Low	~72,511	~99.22%
-1 (Anomalous)	+1 (Normal)	Medium	~281	~0.38%
+1 (Normal)	-1 (Anomalous)	Medium	~90	~0.12%
-1 (Anomalous)	-1 (Anomalous)	High	~470	~0.64%

7.3.4 Risk Classification Output Summary

The final processed output assigns every one of the 73,081 booth records to a risk level. The complete output dataset contains the following columns:

- All original raw columns (State, Constituency, Year, Registered Voters, Votes Polled)
- All five engineered feature columns
- IF_Score (Isolation Forest prediction)
- LOF_Score (Local Outlier Factor prediction)
- Combined_Anomaly_Score (averaged normalized score)
- Risk_Level (Low / Medium / High)
- Anomaly_Reason (human-readable explanation)

This output dataset is the primary deliverable of the anomaly_detection.py module and serves as the data source for all dashboard visualizations.

7.3.5 Filter-Based Output Behavior

One of the most practically useful aspects of the system's output is its responsiveness to user-applied filters. When a user selects specific states or years using the sidebar filters, all outputs update dynamically:

- The summary metric cards reflect counts only for the filtered subset
- The pie chart re-renders with the filtered risk distribution
- The suspicious booth table shows only booths from the selected states/years
- The trend graph and bar charts update to reflect the filtered data

This means the system can effectively serve as both a national-level overview tool and a state-specific or election-specific deep-dive investigation tool — all within the same interface.

Example Filter Scenarios:

Filter Applied	Use Case	Expected Output Change
Select only Bihar	Investigate Bihar-specific patterns	Higher proportion of High Risk booths visible
Select only 1977	Analyze post-Emergency election	Higher average turnout values across all outputs
Select Bihar + 1977	Investigate specific high-risk combination	Concentrated view of top suspicious booths
Select small states only	Check for representation bias	Fewer records, potentially different risk distribution

CHAPTER 8: CONCLUSION

8.1 Merits

The following section details the key strengths and accomplishments of the system across multiple dimensions — technical, functional, design, and academic.

8.1.1 Effective Anomaly Detection with Dual Algorithm Ensemble

One of the most significant technical achievements of this project is the successful implementation and combination of two complementary unsupervised machine learning algorithms — Isolation Forest and Local Outlier Factor — into a coherent ensemble framework.

- Each algorithm independently detects anomalies using fundamentally different mathematical approaches — Isolation Forest through random tree path lengths and LOF through local density comparison.
- By requiring a booth to be flagged by **both** algorithms before being classified as High Risk, the system significantly reduces the probability of false positives.
- The dual-algorithm approach also increases the coverage of different anomaly types — Isolation Forest captures global outliers while LOF captures contextual, locally unusual patterns that a single global algorithm might miss.
- The results validate this design choice — only 0.64% of booths were classified as High Risk, a proportion that is both statistically meaningful and practically manageable for investigation purposes.
- The system successfully identified known historically significant electoral anomalies — such as booths in Bihar, West Bengal, Uttar Pradesh, and Jammu & Kashmir — without being explicitly programmed with knowledge of these historical events. This demonstrates genuine pattern recognition capability rather than rule-based flagging.
- The contamination parameter of 5% was set deliberately to cast a reasonably wide net at the individual model level, while the dual-flag High Risk requirement tightens this to only the most extreme and consistent outliers — a well-balanced design decision.

8.1.2 Meaningful and Contextual Feature Engineering

The quality of a machine learning model is ultimately determined by the quality of the features it operates on. The five engineered features developed for this system represent a carefully considered set of variables that capture multiple distinct dimensions of electoral behavior:

- **Turnout Percentage** provides the absolute measure of participation at each booth — the foundational variable from which all other features are derived.

- **Turnout Change** captures the temporal dimension — measuring how a booth's participation rate has shifted between consecutive election cycles. This feature is critical for identifying booths that show sudden, unexplained jumps in turnout.
- **State Average Turnout** establishes a regional benchmark that accounts for the natural geographic variation in Indian electoral participation across different states.
- **State Deviation** directly quantifies how far each booth deviates from its regional norm — the single most powerful predictor in the system, as evidenced by its dominance in the anomaly reason outputs.
- **Normalized Turnout** ensures mathematical fairness across features with different numerical scales, preventing any single feature from dominating the distance calculations in LOF.

Together, these five features enable the system to make contextually informed judgments. A booth is evaluated not just against a global threshold but against its own historical behavior and its regional peers. This contextual intelligence is what distinguishes this system from naive threshold-based approaches and makes it genuinely useful for real-world electoral monitoring.

8.1.3 Scalability and Computational Efficiency

The system successfully processed **73,081 booth records** spanning 11 election cycles across 43 states — a substantial dataset — within reasonable computational time on a standard personal computer.

- Isolation Forest's $O(n \log n)$ time complexity ensures that it scales well even as the dataset grows larger.
- The Pandas-based preprocessing pipeline is memory-efficient and handles the full dataset without requiring distributed computing infrastructure.
- The modular architecture — with preprocessing, feature engineering, and anomaly detection in separate Python modules — means that individual components can be updated or replaced without rebuilding the entire pipeline from scratch.
- The Streamlit dashboard loads and renders all visualizations quickly, with interactive filter updates completing in near real-time even for the full 73,081-record dataset.
- This scalability means the system could be extended to even larger datasets — for example, incorporating state assembly election data or booth-level data from future election cycles — without requiring fundamental architectural changes.

8.1.4 User-Friendly and Accessible Dashboard Interface

A technically sophisticated system that cannot be used by its intended audience has limited practical value. The Streamlit dashboard specifically addresses the accessibility challenge that has historically limited the adoption of computational electoral monitoring tools.

- The dashboard requires **no programming knowledge** to operate — any user who can use a web browser can apply filters, explore visualizations, and interpret results.

- The three-tab structure — Overview, Suspicious Booths, and Data Analysis — organizes information logically, from high-level summary to specific actionable outputs to exploratory analysis.
- The **Anomaly Reason** column in the suspicious booth tables provides plain-language explanations for why each booth was flagged, removing the need for users to interpret raw numerical anomaly scores.
- The **interactive filters** for state and year allow users to focus the analysis on specific regions or election cycles — making the tool useful for both national-level oversight and targeted regional investigations.
- The **download functionality** on the suspicious booth table allows users to export results as CSV for further offline analysis or formal reporting.
- The color-coded risk levels — green for Low, orange for Medium, red for High — provide immediate visual intuition about the severity of detected anomalies without requiring any technical interpretation.
- The system effectively serves multiple user types simultaneously — a senior official who needs a quick summary can use the Overview tab, an investigator who needs specific booth details can use the Suspicious Booths tab, and a researcher who wants to explore data patterns can use the Data Analysis tab.

8.1.5 Historically Validated Outputs

A key merit of the system's outputs is that they align well with independently documented historical knowledge about Indian electoral integrity:

- The **Bihar dominance** in the High Risk list aligns with Bihar's well-documented history of electoral irregularities, including booth capturing and turnout manipulation — particularly in elections from the 1970s through the 1990s.
- The **1977 election peak** in the turnout trend graph matches the historically recorded extraordinary voter mobilization of the first post-Emergency election — demonstrating that the system correctly identifies nationally significant electoral events without being explicitly informed about them.
- The **West Bengal — Arambagh 2004** flagging aligns with documented concerns about electoral practices in West Bengal during the Left Front era of that period.
- The **Rae Bareilly 1984** flagging is historically noteworthy given that constituency's political significance during that election cycle.
- The **Jammu & Kashmir — Ladakh** entries across 1989 and 1996 reflect the historically complex and often disputed nature of elections in that sensitive region.
- The fact that the system produces outputs consistent with historical knowledge — without being explicitly programmed with that knowledge — is strong evidence that the anomaly detection approach is capturing genuine statistical signals rather than random noise or artifacts of the data.

8.1.6 Modular and Maintainable Code Architecture

The system is implemented across three clearly separated Python modules, each with a well-defined single responsibility:

- `data_preprocessing.py` — handles all data cleaning, validation, and preparation
- `anomaly_detection.py` — handles all machine learning model training, prediction, and risk classification
- `app.py` — handles all dashboard visualization and user interaction logic

This separation of concerns provides several important practical benefits:

- Each module can be tested, debugged, and updated independently without affecting the others.
- New preprocessing steps can be added to `data_preprocessing.py` without modifying the detection logic.
- The anomaly detection algorithms in `anomaly_detection.py` can be swapped or extended with new models without touching the dashboard code.
- The dashboard in `app.py` can be redesigned or extended with new visualizations without altering the underlying analytical pipeline.
- The modular structure also makes the codebase significantly more readable and understandable for future developers who may wish to extend or adapt the system.

8.1.7 Contribution to Electoral Accountability

Beyond its technical merits, the project makes a meaningful contribution to the broader goal of democratic accountability:

- It demonstrates that sophisticated anomaly detection can be applied to Indian electoral data using entirely open-source tools — Python, Scikit-learn, Pandas, and Streamlit — making the approach replicable by anyone without requiring proprietary software or expensive infrastructure.
- It provides a fully reproducible and transparent methodology — anyone with the same dataset and the same code will produce identical outputs, enabling independent verification of results.
- It surfaces specific, named constituencies and election years for potential investigation — moving from abstract statistical concern to concrete, investigable cases with supporting evidence.
- It establishes a working template that could be adopted and adapted by election monitoring organizations, academic researchers, investigative journalists, and civil society groups working on electoral integrity across India and other large democracies.
- It demonstrates the practical feasibility of machine learning-based electoral monitoring at scale — contributing to the growing body of evidence that computational methods can and should play a larger role in democratic oversight.

8.2 Limitations

An honest and thorough acknowledgment of limitations is as important as documenting strengths. The following limitations should be carefully considered by anyone using or building upon this system.

8.2.1 Absence of Ground Truth Labels

The most fundamental limitation of this system — and of all unsupervised anomaly detection approaches applied to electoral data — is the complete **absence of verified ground truth labels**.

- The system identifies statistical anomalies, not proven instances of electoral fraud. A booth flagged as High Risk may have a legitimate, non-fraudulent explanation for its unusual turnout figures — such as a highly contested local election, a major political event in the region, or a demographic shift in the constituency.
- There is no labeled dataset available that definitively marks specific booths as fraudulent or clean, which means it is impossible to compute traditional supervised learning evaluation metrics such as precision, recall, F1-score, or ROC-AUC curve for the system's outputs.
- Without ground truth, the system cannot be formally validated in the conventional machine learning sense. Its validation is limited to indirect methods — such as checking whether flagged booths align with historically documented irregularities, as was done in Chapter 6.
- This is a limitation of the problem domain itself rather than this specific implementation, and it is widely acknowledged across the electoral fraud detection literature. It underscores the importance of treating the system's outputs as investigative leads rather than definitive verdicts.

8.2.2 Constituency-Level Rather Than True Booth-Level Granularity

- The dataset available for this project contains **constituency-level aggregated data** rather than true **individual polling booth-level data**. Each record represents the total votes polled across all booths within a parliamentary constituency, not the specific votes at a single polling station.
- This means the system cannot identify irregularities at the level of individual polling stations within a constituency — it can only flag the constituency as a whole.
- True booth-level data, where available, would enable significantly more precise and actionable anomaly detection — narrowing suspicion to specific booths within a constituency rather than flagging the entire constituency.
- Future work should prioritize obtaining genuine booth-level data from the Election Commission of India or other reliable sources to enable this finer-grained analysis.

8.2.3 Unsupervised Learning — No Performance Guarantee

- Because the system uses unsupervised learning algorithms, there is no mathematical guarantee of detection performance in the absence of labeled data.
- The contamination parameter — set at 5% for both Isolation Forest and LOF — was chosen based on reasonable assumptions about the expected proportion of anomalies in the dataset, but this value is inherently arbitrary. A different contamination setting would produce a different number of flagged booths.
- The models have no way of knowing whether a flagged booth is genuinely suspicious or whether an unflagged booth contains subtle manipulation that the chosen features did not capture.
- The risk of **false negatives** — genuine irregularities that the system fails to detect — cannot be quantified without ground truth labels. There may be forms of electoral manipulation that do not manifest as unusual turnout figures and would therefore be invisible to this system entirely.

8.2.4 Limited Feature Set

- While the five engineered features capture important dimensions of electoral behavior, they represent only a subset of the information that could potentially be useful for anomaly detection.
- Features not currently included in the model — such as **margin of victory**, **number of candidates**, **voter demographics**, **geographic remoteness**, **presence of security forces**, and **historical complaint records** — could significantly improve detection accuracy if incorporated.
- The current feature set is entirely turnout-based. A sophisticated form of electoral manipulation that maintains plausible turnout figures while manipulating vote shares between candidates would not be captured by any of the current features.
- The absence of demographic data — such as literacy rates, urbanization level, and economic indicators — means the system cannot distinguish between unusual turnout driven by genuine civic enthusiasm and unusual turnout driven by manipulation.

8.2.5 Historical Data Only — No Real-Time Capability

- The system is designed to operate on historical, pre-collected datasets. It does not have the capability to ingest and analyze live data streams during an ongoing election.
- Real-time anomaly detection — where suspicious patterns are flagged as votes are being counted — would require a fundamentally different architecture involving streaming data pipelines, real-time database connections, and automated alerting systems.
- The current system is therefore a **post-hoc analytical tool** rather than a live monitoring system. Its findings can inform future election planning and policy but cannot directly intervene in an ongoing election process.
- Developing real-time capabilities would be a significant and valuable area of future work, as described in Chapter 9.

8.2.6 Potential for Misuse and Misinterpretation

- A system that flags specific constituencies and states as "High Risk" carries a significant risk of misuse if its outputs are communicated irresponsibly.
- Without proper contextual interpretation, the outputs of this system could be used to make unfounded allegations of fraud against political parties, candidates, or regional administrations — causing reputational harm without due process.
- The system explicitly does not determine whether fraud occurred — it identifies statistical anomalies that warrant investigation. This distinction must be clearly communicated to any user of the system, and the documentation repeatedly emphasizes this point.
- There is also a risk of **confirmation bias** — users who already suspect a particular region or political party of fraud may use the system's outputs to confirm their prior beliefs rather than approaching them as neutral statistical findings.
- Responsible deployment of this system requires clear communication of its limitations, the provisional nature of its outputs, and the necessity of human investigation before any conclusions are drawn.

8.2.7 Geographic Representation Imbalance

- The dataset is heavily skewed toward large northern Indian states, particularly Uttar Pradesh and Bihar, which together account for a disproportionate share of total records.
- This imbalance means that the anomaly detection models are trained primarily on the distributional characteristics of large-state data. Booths from smaller, less-represented states and union territories may be evaluated against standards that do not accurately reflect their regional context.
- The LOF algorithm is particularly susceptible to this issue — its local density estimates depend on having a sufficient number of similar neighbors, which may not be available for booths from underrepresented regions.
- Future work should explore stratified sampling or region-specific model training to address this geographic imbalance.

8.2.8 No Integration with External Complaint or Legal Records

- The system operates exclusively on quantitative electoral data and has no integration with existing complaint records, court orders, election observer reports, or police records related to electoral violations.
- Integrating such qualitative and legal data sources could significantly enhance the system's ability to validate its outputs and prioritize cases for investigation.
- A constituency flagged as High Risk by the model AND with an existing record of electoral complaints would represent a much stronger case for investigation than a purely statistically flagged constituency with no corroborating evidence.
- Building such an integration would require access to structured complaint and legal databases, which are not currently publicly available in a digitized, machine-readable format.

8.3 Overall Conclusion

The Machine Learning-Based Election Irregularity Detection System represents a meaningful and technically sound contribution to the field of computational electoral monitoring. By combining two complementary anomaly detection algorithms, carefully engineered contextual features, and an accessible interactive dashboard, the system successfully identifies statistically unusual polling booths from a large historical dataset in a way that is consistent with independently documented electoral history.

The system is not a perfect fraud detector — no such tool exists, and the limitations documented in Section 8.2 must be taken seriously by anyone using or building upon this work. It is best understood as an intelligent triage tool — one that directs the attention of human investigators toward the small fraction of cases most likely to warrant scrutiny, making electoral oversight more efficient, consistent, and evidence-based.

The project demonstrates that open-source machine learning tools, applied thoughtfully to publicly available electoral data, can produce meaningful analytical outputs with real democratic value. As electoral data becomes increasingly available in digitized form, and as machine learning capabilities continue to advance, systems like this one have the potential to play an increasingly important role in protecting the integrity of democratic processes — not by replacing human judgment but by informing and augmenting it.

CHAPTER 9: FUTURE SCOPE

9.1 Introduction

The Machine Learning-Based Election Irregularity Detection System developed in this project represents a foundational implementation of computational electoral monitoring. While the system successfully achieves its defined objectives — processing large-scale historical election data, detecting statistical anomalies, classifying booths by risk level, and presenting findings through an interactive dashboard — it also opens up a wide range of possibilities for future enhancement, extension, and deployment. This chapter systematically explores these possibilities, organized thematically across technical, functional, data, and deployment dimensions.

Each future scope item described in this chapter addresses either a specific limitation identified in Chapter 8 or an opportunity to extend the system's capabilities beyond its current boundaries. Together, these directions form a comprehensive roadmap for the next generation of this system.

9.2 True Booth-Level Data Integration

Current State:

The present system operates on constituency-level aggregated data, where each record represents the total electoral activity across all polling booths within a parliamentary constituency.

Future Direction:

- The most impactful near-term improvement would be the integration of **genuine polling booth-level data** — where each record represents a single physical polling station with its own registered voter count, votes polled, and turnout figure.
- The Election Commission of India publishes booth-level data for many recent elections, and organizations such as the Trivedi Centre for Political Data (TCPD) have been working toward making this data more accessible.
- With true booth-level data, the system's granularity would improve dramatically — instead of flagging an entire constituency, it could pinpoint the specific booth or booths within a constituency that show anomalous behavior.
- This would make the system significantly more actionable for election monitoring authorities, who could dispatch physical observers or request re-polling specifically at flagged booths rather than entire constituencies.
- The analytical pipeline, feature engineering logic, and dashboard architecture developed in this project are all compatible with booth-level data and would require only minor modifications to accommodate the finer granularity.

9.3 Real-Time Election Monitoring System

Current State:

The current system is a batch-processing, post-hoc analytical tool. It analyzes historical data after elections have concluded and produces static outputs that cannot influence ongoing electoral processes.

Future Direction:

- A natural and highly valuable extension would be the development of a **real-time election monitoring capability** that can process and analyze data as it is generated during an ongoing election.
- This would require the development of a **streaming data pipeline** — using technologies such as Apache Kafka or Python's asyncio framework — capable of ingesting live data feeds from polling stations as votes are counted.
- Real-time anomaly detection would involve training the models on historical data and then applying them to incoming live data points, flagging suspicious patterns as they emerge rather than after the fact.
- An **automated alerting system** could be integrated — sending notifications to designated officials or observers when a booth's real-time turnout crosses a statistically significant threshold relative to its historical norm.
- The dashboard could be extended with a **live monitoring panel** showing real-time risk level updates as data comes in from across the country.
- While this represents a significant architectural expansion beyond the current system, the core algorithmic components — Isolation Forest, LOF, and the feature engineering logic — are all compatible with a streaming implementation.

9.4 Incorporation of Additional Features and Data Sources

Current State:

The system currently uses five features derived entirely from turnout data — Turnout Percentage, Turnout Change, State Average Turnout, State Deviation, and Normalized Turnout.

Future Direction:

Incorporating a richer set of features from diverse data sources would significantly improve the system's detection accuracy and reduce both false positives and false negatives. The following additional features are candidates for inclusion:

Electoral Features:

- **Margin of Victory:** Constituencies where the winning margin is very small are more likely to be the target of deliberate manipulation — this feature would help the system prioritize high-stakes races.
- **Number of Candidates:** A very high number of candidates may fragment the vote in unusual ways; a very low number may indicate suppression of competition.
- **Vote Share Concentration:** The proportion of votes going to the top candidate — extremely high concentrations may indicate ballot stuffing in favor of a specific party.
- **Postal Ballot Percentage:** Unusually high proportions of postal ballots relative to in-person votes could indicate specific forms of manipulation.
- **Re-polling History:** A binary flag indicating whether a constituency has previously been subject to a re-polling order — which would indicate prior documented irregularity.

Demographic and Socioeconomic Features:

- **Literacy Rate:** Lower literacy areas may be more susceptible to certain forms of voter coercion or impersonation.
- **Urbanization Level:** Urban and rural booths have very different baseline turnout patterns — incorporating this feature would improve the State Deviation calculation.
- **Population Density:** Dense urban constituencies behave differently from sparse rural ones.
- **Economic Development Index:** Economically underdeveloped regions may show different patterns of electoral participation and vulnerability to vote-buying.

Geographic Features:

- **Distance from District Headquarters:** Remote booths may be harder to monitor and more susceptible to irregularities.
- **Terrain Classification:** Mountainous, coastal, forest, or plain terrain affects booth accessibility and therefore expected turnout.
- **Constituency Sensitivity Classification:** The ECI already classifies certain constituencies as "sensitive" or "hyper-sensitive" — incorporating this classification as a feature would provide valuable institutional knowledge.

9.5 Advanced Machine Learning Techniques

Current State:

The system uses Isolation Forest and Local Outlier Factor — both well-established unsupervised anomaly detection algorithms that operate on static feature vectors.

Future Direction:

Several advanced machine learning techniques could be explored to improve detection performance:

Supervised Learning with Labeled Data:

- If a labeled dataset of confirmed fraudulent and clean booths becomes available — through court records, election tribunal orders, or expert annotation — the problem could be reframed as a **supervised binary classification** task.
- Algorithms such as **Random Forest**, **Gradient Boosting (XGBoost, LightGBM)**, or **Support Vector Machines** could be trained on labeled data and would likely achieve significantly higher precision and recall than unsupervised methods.
- Even a partially labeled dataset could be used in a **semi-supervised learning** framework — using the labeled examples to guide the unsupervised detection of unlabeled anomalies.

Deep Learning Approaches:

- **Autoencoders** — neural networks trained to reconstruct their input — are a powerful deep learning approach to anomaly detection. A booth whose features are poorly reconstructed by an autoencoder trained on normal booths is likely anomalous. This approach could capture complex, non-linear relationships between features that Isolation Forest and LOF miss.
- **Long Short-Term Memory (LSTM) networks** could be used for time-series anomaly detection — learning the temporal patterns of each booth's turnout history and flagging deviations from the learned pattern.
- **Graph Neural Networks (GNNs)** could model the spatial relationships between adjacent constituencies — detecting anomalies that cluster geographically in ways that suggest coordinated manipulation.

Ensemble Expansion:

- The current two-algorithm ensemble (IF + LOF) could be expanded to include additional algorithms such as **One-Class SVM**, **Elliptic Envelope**, or **DBSCAN-based outlier detection**.
- A **weighted voting ensemble** — where each algorithm's vote is weighted by its historical performance — could produce more stable and reliable classifications than the current equal-weight approach.

9.6 Geographic Information System (GIS) Integration

Current State:

The current dashboard presents state-level and constituency-level information in tabular and chart form but has no geographic mapping capability.

Future Direction:

- Integrating a **Geographic Information System (GIS)** component would add a powerful spatial dimension to the system's outputs.

- A **choropleth map of India** — where constituencies are color-coded by risk level — would allow users to immediately identify geographic clusters of high-risk booths, which may indicate coordinated, region-wide manipulation rather than isolated incidents.
- Libraries such as **Folium**, **GeoPandas**, or **Plotly's Mapbox integration** could be used to implement interactive maps within the existing Streamlit dashboard.
- Spatial clustering analysis — identifying whether high-risk booths are geographically concentrated or randomly distributed — would add an important additional dimension to the anomaly interpretation.
- A constituency-level map with drill-down capability — clicking on a constituency to see its detailed risk profile and historical trend — would significantly enhance the investigative utility of the dashboard.
- GIS integration would also make the system's outputs more intuitive for non-technical stakeholders such as election observers and journalists, for whom maps are often more immediately comprehensible than tables and charts.

9.7 Natural Language Processing for Complaint Integration

Current State:

The system operates exclusively on structured quantitative data and has no capability to process or incorporate unstructured text data such as electoral complaints, observer reports, or news articles.

Future Direction:

- **Natural Language Processing (NLP)** techniques could be used to automatically process and analyze textual data sources related to electoral integrity.
- A **complaint analysis module** could process the text of electoral complaints filed with the Election Commission, extracting constituency names, election years, and types of alleged irregularities, and cross-referencing these with the system's anomaly detection outputs.
- **Sentiment analysis** of news articles and social media posts during election periods could provide real-time indicators of electoral tension or reported irregularities at specific locations.
- **Named Entity Recognition (NER)** could automatically extract constituency names, candidate names, and dates from large volumes of unstructured text, enabling automatic population of a complaint database.
- Integrating NLP-derived signals with the quantitative anomaly scores could produce a richer, multi-modal risk assessment — combining statistical evidence from voting data with textual evidence from complaints and reports.

9.8 Comparative International Application

Current State:

The system is currently designed and calibrated specifically for Indian electoral data, using Indian administrative boundaries and the Indian electoral calendar.

Future Direction:

- The core methodology of the system — feature engineering on turnout data, dual-algorithm anomaly detection, and interactive dashboard visualization — is **generalizable to electoral data from any country** with publicly available polling station-level data.
- Adapting the system for other large democracies — such as **Brazil, Indonesia, Nigeria, Pakistan, Bangladesh**, or countries in Eastern Europe — would require only modifications to the data ingestion and feature engineering modules to accommodate different administrative structures and data formats.
- Comparative international application would also enable **cross-country benchmarking** — identifying whether certain patterns of electoral anomaly are unique to specific countries or represent universal signatures of manipulation across different political and institutional contexts.
- International election monitoring organizations — such as the Carter Center, the EU Election Observation Mission, and the OSCE/ODIHR — could potentially adopt and adapt this system for their observation missions in countries with available digital electoral data.

9.9 Automated Report Generation

Current State:

The system produces interactive dashboard outputs that require a user to actively explore and interpret the results. There is no automated mechanism for generating formal, structured reports suitable for official submission.

Future Direction:

- A **PDF report generation module** could be developed that automatically compiles the system's outputs — summary statistics, risk distribution charts, suspicious booth tables, and trend graphs — into a formatted, printable report.
- Libraries such as **ReportLab** or **WeasyPrint** could be used to generate professional PDF reports directly from the dashboard with a single button click.
- The report could be customized based on the user's active filter selections — generating a state-specific or year-specific report as needed.
- **Automated narrative generation** using large language models (LLMs) could produce plain-language summaries of the key findings for each report — describing

the most suspicious booths, the overall risk distribution, and notable trends in language accessible to non-technical readers.

- This capability would make the system directly usable for formal election observation missions, where observers are required to submit structured written reports within tight deadlines.

9.10 Integration with Official Election Commission Systems

Current State:

The system is a standalone analytical tool that operates independently of official election management systems.

Future Direction:

- In the long term, the most impactful deployment scenario would be **direct integration with the Election Commission of India's data systems**.
- If the system were connected to the ECI's official data infrastructure, it could receive verified booth-level data automatically upon the conclusion of polling and produce anomaly detection outputs within hours of result declaration.
- This would enable **election officials** to use the system's outputs as part of the official post-election review process — systematically examining flagged booths before results are formally certified.
- Integration with the ECI's **ENCORE (Election Commission Result) system** or its **Voter Turnout Application** — which already collects real-time turnout data — would be a natural starting point for this integration.
- Such official integration would also address the ground truth problem identified in Chapter 8 — over time, the system's outputs could be validated against the ECI's own investigation records, enabling the development of a labeled dataset for supervised learning.

9.11 Mobile Application Development

Current State:

The system is accessible as a web-based Streamlit dashboard, which requires a desktop or laptop browser for optimal use.

Future Direction:

- Developing a **mobile application** version of the dashboard would significantly extend accessibility — election observers deployed in the field could access risk assessments and suspicious booth lists directly from their smartphones.

- A mobile app could incorporate **push notifications** — alerting observers when new high-risk booths are detected in their assigned region as results come in.
- **Offline capability** — where the app can function without a continuous internet connection and sync when connectivity is restored — would be particularly valuable for observers deployed in remote constituencies with unreliable network access.
- Frameworks such as **React Native** or **Flutter** could be used to develop cross-platform mobile apps that consume the system's analytical outputs via a REST API.

9.12 Continuous Learning and Model Updating

Current State:

The anomaly detection models are trained once on the available historical dataset and do not update as new election data becomes available.

Future Direction:

- Implementing a **continuous learning pipeline** would allow the models to retrain automatically as new election data is added to the dataset — ensuring that the system's understanding of "normal" electoral behavior stays current.
- After each new election cycle, the new data could be automatically preprocessed, features engineered, and the models retrained — with the updated models deployed to the dashboard without manual intervention.
- **Model versioning** — using tools such as MLflow or DVC (Data Version Control) — would allow different versions of the trained models to be tracked, compared, and rolled back if necessary.
- Over time, the accumulation of more election cycles would improve the statistical power of the models, particularly for detecting subtle, low-magnitude anomalies that are difficult to identify with limited historical data.

9.13 Conclusion of Future Scope

The future directions outlined in this chapter collectively point toward a vision of the Election Irregularity Detection System as not merely an academic project but a genuinely deployable tool for democratic accountability. The path from the current implementation to that vision involves advances in data availability, computational infrastructure, institutional partnerships, and technical sophistication — none of which are insurmountable given sufficient time and resources.

The most immediately actionable improvements — true booth-level data integration, additional feature engineering, and GIS mapping — could be implemented as direct extensions of the current codebase with relatively modest effort. The more ambitious directions — real-time monitoring, deep learning models, and official ECI integration — represent longer-term research and development goals that would require collaboration between data scientists, election administrators, and policy makers.

What this project has established is the foundational proof of concept: that machine learning-based electoral anomaly detection is technically feasible, produces historically consistent outputs, and can be made accessible to non-technical users through thoughtful dashboard design. Every future direction described in this chapter builds on that foundation.

CHAPTER 10: REFERENCES

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